

KARL FRISTON - INTELLIGENCE 3.0

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hello everyone I'm here to talk to you about a vision of artificial intelligence that goes beyond machines and algorithms a vision which Embraces humans and nature as integral parts of a cyber physical ecosystem of intelligence a vision that is based on first principles derived from physics and biology Professor Carl friston just dropped a paper called designing ecosystems of intelligence from first principles now one of the principles is active inference a formulation of adaptive Behavior which can be read as a physics of intelligence active inference says that intelligent systems are those which can accumulate evidence for a generative model of their sensed World in other words they can learn from their observations and update their beliefs accordingly they can also act on their environment to reduce uncertainty and achieve their goals but active inference is not about individual agents it also explains how ensembles of Agents can share beliefs and cooperate through communication protocols this leads to a formal account of collective intelligence that rests on shared narratives and goals so how do we realize this Vision friston proposes a research agenda for the next decade and Beyond which aims to design such ecosystems of intelligence from scratch they suggest developing a shared hyperspatial modeling language and transaction protocols as well as novel methods for measuring and optimizing collective intelligence so why should we care about this Vision it's because it offers a way to harness the power of artificial intelligence for the common good without compromising human dignity or autonomy and because it challenges us to rethink our relationship with technology nature and each other and because it invites us to join in a global community of sense makers who are curious about the world and eager to improve it enjoy the show our shared human journey is filled with examples of simple ideas that were nonetheless hard to discover and some that even once explained remain hard to comprehend their subtle Simplicity belies their far-reaching and deep consequences examples might include the principle of relativity quantum mechanics the principle of parsimony and entropy I think the free energy principle is another example profound and far-reaching yet belied by its simplicity it has on the one hand been dismissed as a triviality or even tautology and on the other hailed

as revolutionary and everything in between it has the potential to reshape how we view the connection between inanimate matter and the living things and even to answer the question of how and why Consciousness and intelligence might emerge from physical matter and processes that means a world where all things from particles to people to the largest systems all move or evolve according to two which combine the first is a smooth flowing Evolution think of planets orbiting a star or waves Rippling over water or through Quantum fields the second is a random process that knocks around the smooth flow in unpredictable ways think of twinkling stars or audio static you might think of these two processes as a kind of Order and Chaos or yin and yang forever intertwined and enmeshed these two very different effects combine into a chaotic flow which may be entangled in a kind of tropical storm while still maintaining a semblance of structure and things think about raindrops undulating down to earth in a state of constant flux yet still droplets or a human being growing and adapting to the surprises of Life all the while remaining an individual physical entity biology often asks the question what must things do in order to exist Professor Friston has turned that question on its head and developed perhaps the ultimate existential formalism he asks if things exist what must they do from the foundation of stochastic differential equations Friston demonstrates that things which are defined by a Markov blanket must always move towards a pullback attractor a special set of attracting states which maintains the Integrity of the Markov blanket and therefore a thing's coherence and identity over time one of the profound consequences of this is that the Dynamics of such a system its laws of motion will manifest a form of flow Dynamics which can be interpreted as Bayesian active inference in other words such a thing maintains an internal equivalent of a generative model encoding beliefs about the world and itself the thing uses the model to decide actions and then performs a Bayesian update based on the outcomes for me the idea that inference something widely perceived as purely abstract or mathematical the idea that it can be driven by simple Laws of Motion dynamically maintaining the boundaries between things maintaining order in the face of Chaos is frankly astonishing what's more the free energy principle is so General that it applies that all scales of size and time leading to an ecosystem of things interacting across scales perhaps in that multi-scale active inference we might finally find the keys to a mathematics of emergence and consciousness this episode is sponsored by Numerai I'm extremely grateful to them for sponsoring machine learning Street talk I mean remember I do everything myself on this channel it's a lot of time it's a lot of hard work it's very expensive I pay for other software licenses and all of the equipment so um yeah having that support from Numerai means a lot to me and it means that I

can keep doing what I'm doing basically so thank you to numerai so a little bit about numerai they are a data science competition platform to predict the stock market they've already paid out over 50 million uh dollars for 5 000 models on their platform and they say that it's the highest paying data science platform in the world now you can get started really easily for free and they've got a couple of examples on their GitHub repo actually using xgboost um and the data comes in parquet format so you can get up and running really easily you can submit your predictions weekly or daily you can do it either manually or you can do it automatically and they provide many many years of of back testing data as well so you can fine tune and test your models and do a bunch of statistical Diagnostics you can work your way up the leaderboard for bragging rights and stake your model with their NMR cryptocurrency to earn rewards now staking is a vote of confidence of your model and the reason they do that is to prevent overfitting and to reduce the um you know bad models from kind of contributing bad intelligence to their to their aggregate now a well-performing model is rewarded in proportion to its stake but a poorly performing model is punished by burning a portion of these stake now numeri have a huge community of data scientists of all levels swapping ideas and advice and they have some Community forums as well which you should join now um just a personal note from me this is a form of betting on the predicted performance of your models which can go up or down remember the data is abstracted from the actual performance of the stocks they they don't correlate to actual stock performance so there are folks on the platform who have made lots of money but there's also folks who have lost money as well so please be responsible and only stake what you can afford to lose and have confidence in try to have fun use it as a place to sharpen up your data science skills and to be the best version of yourself and use all of the latest models that we've been talking about here on on Street Talk but um yeah anyway thank you again to numerai for sponsoring us Professor frisson it's an absolute honor to meet you well thank you very much for having me um so we've had you on the show two times now and in the first show we went into um Exquisite detail about the free energy principle and active inference so um you're extremely famous for introducing this existential imperative which is to say if things survive what must they do they gain information about the world around them in a cybernetic Loop we find a model which fits well while maintaining High entropy if you have a higher entropy model you have greater flexibility to adapt to new information it's an absolutely beautiful idea so um welcome thank you nicely articulated I was speaking with Keith yesterday and he said out of all of the guests that we've had on m1st you are by far his favorite and he says he looks up to you very much so he's he's very gutted that he

couldn't come today to be part of this it's very gracious of it so um you just wrote a paper and it's called a designing ecosystems of you led with this white paper lays out a vision of research and development in the field of artificial intelligence for the next decade and Beyond its denouement I wasn't sure if I was going to say that way is a cyber physical ecosystem of natural and synthetic sense making in which humans are integral participants what we call shared intelligence this vision is premised on active inference the formulation of adaptive behavior that can be read as the physics of intelligence and which inherits from the physics of self-organization so um you went on and by the way there's an interesting link with Michael Levin's work here so we had him on the show recently and maybe even transhumanism as well we could we can get into that um but you went on you said in this context we understand intelligence as the capacity to accumulate evidence for a generative model of one's sensed World also known as self-evidencing over multiple scales and crucially you said active inference foregrounds and existential imperative of intelligent systems namely curiosity or the resolution of uncertainty and the same imperative underwrites belief sharing in ensembles of agents in which certain aspects which is to say factors of each agent's generative model provide common ground or a frame of reference so can you sketch some of this out for me yes you've used all my favorite words though I presume that I actually used all the wear fat to be truthful um Maxwell ramstead was with one of the key architect of this so this was a um this white paper was a response to a brief just to think seriously and pragmatically how this sort of high Church theoretical approach to you know intelligence and self-organization would play out in industry in the way that we actually use um technology um over the next decade and that white paper is the product of our massive Asians and discussions um but just drilling down on on the sort of basic message there um the emphasis was as you say on um what it is to exist and how that would manifest in terms of intelligent behavior um and to a certain extent certainly I and I think a number of the other co-authors were reading intelligence as the kind of inference that you would need to do in order to maintain your existence so hence the existential imperative I have to say that's a slightly um poetic interpretation of free energy principle which is the other way around of course that if you exist it looks as if you are behaving intelligently if you read intelligence as the right kind of belief updating that enables you to maintain yourself in some characteristic state so that's that's essentially um what that existential imperative was all about you just a one way of reading the physics the mechanics of systems that are open in open exchange with the rest of their world or their Eco Niche um simply in virtue of the fact that they are around for an extended periods of period of time and

have this characteristic set of states that they occupy that defines them as the kind of thing that they are I think the move really that was inspired by this remit to look okay well that's very nice you can write lots of papers about that and have a nice chats with your friends and colleagues about it but how does that actually work in practice and of course in practice you have to think about where you're going to deploy this um perspective deploy the kind of technology that would you know um ensue from that perspective of course we're talking about communication we're talking about a universal world of lots of creatures like me and you so now we get to um the next level of application of these existential imperatives in the context of not just intelligent behavior and just Open brackets Behavior here is quite important you know often I hear um underwriting the move from AI artificial intelligence to IA where it's intelligent age artifacts with agency that do have behaviors that have a centered kind of behavior and or at least an intelligent uh kind of behavior close brackets um so um we're now moving from um thinking about a single thing a particle or a person or a computer and thinking now about having lots of things that are talking to each other and they're coexisting so now we're in the world of distributed intelligence distributed cognition um one thing which um you know you mentioned explicitly this notion of belief for sharing so now we're in a different kind of world where we're thinking okay we think we understand the imperatives for good behavior the necessary kinds of behaviors that you know you you identity highlighted curiosity I think is absolutely Central in terms of you know whole marketing and character drives and the kinds of behaviors that you and I engage with are truly intelligent or sentient Behavior would entail but now the idea is well what would that look like and um when you've got lots of these things talking to each other well we know what it looks like we talk to each other literally so it's all about language it's all about uh belief sharing shared intelligence distributed cognition would be one thing you might find in the in the Life Sciences um if you were a computer scientist it's all about sort of the right kind of message passing on very very large Factor graphs um if you're if you're a linguist or into uh Evolution and psychology it's all about neat cultural Niche construction the emergence of language and so I found that quite fascinating as something which you know you really had to get on top of to think well what next and what kind of Technologies would you need to realize and what would they look like you know one way that's been framed to me by my new friends now in Industry um is you know what would what's it going to look like what what is um AI or now IA going to look like um in the future taking the kind of perspective you know over the decades or indeed centuries where you're moving from sort of the Industrial Age to the age of information and I think the

notion now is we're now come to the end of the age of information now we're in the age of intelligence and think about what that might look like and how would one equip people with the right kind of technology and infrastructure um to you know to realize the potential of shared intelligence essentially yeah that's very interesting so we spoke with Luciano Floridi at Oxford and he's invented this philosophy of information he talks about the infosphere and how we're we're out we now have third order technology essentially so things like Amazon and Facebook and he talks about the um diminishment of our ontology and agency in the info sphere and using this information as a substrate thing I think is very interesting because that's kind of alluding to what you're going to but um you said something actually in your paper which gave me pause for thought you said the AI age may end up being a distributed network of intelligent systems which interact frictionally in friction frictionlessly in real time and composing to emergent forms of intelligence at subordinate scales as you were just alluding to and the nodes of such a distributed interconnected ecosystem May then be human users as well as human designed artifacts that embody or Implement forms of intelligence now this really did make me wonder because is I think we're already there now and uh so when when we had Floridian he was talking about how our digital identity has already distributed we were consumers and then we became barcodes and now there's this kind of infinite fractionation in the info sphere and it's almost as if Humanity isn't the substrate anymore information is the substrate and he said you can devalue human skills you can remove remove human responsibility you can reduce human control you can erode human self-determination but the rock that thought that little drop of water was nothing 18 years later has a hole in it because drop after drop after drop the drop Will shape the stone will shake the Rock and I think what he meant by that is being ensconced and enmeshed in technology to such a high degree and information being the first class citizen in our society is kind of truncating our very existence I mean what would you say to that beautifully effectively expressed very poetic I'd have to think about that very carefully I'm sure there are deep truths there um and certainly from uh from the point of view of a physicist your information is obviously um the thing it is it is uh in many respects the you know the substance of our universe and our existence and you know also acknowledging that energy is information and energy is just a potential and so we're all about we're all in the game of realizing Potentials in a sort of folk psychology sense but also literally in terms of minimizing certain potentials for example self-information uh would be the the free energy potential that you're that you're minimizing the surprise and again coming back to this sort of making you know belief sharing in the service of minimizing uh minimizing a

surprise and realizing our our potential um so I but I don't see that as a negative thing I think that's just an acknowledgment that you know at the end of the day um the way that we model and articulate and talk about um ourselves our lived World whether we're a Quantum physicistic um a statistical physicist or a classical physicist or I have an interest in our in the equivalent mechanics at each of those levels it's all about information it's all about the probability of being in this particular State you just take the um the negative log of that that is the self-information of this particular state so it's all you know if you look at quantum physics you know what else is there it boils down to um and it's sort of most Elemental and scale free or background free form and Quantum information Theory and so if you can reduce it to you know to Quantum information Theory which many people including friends of Mike Levin and think one can then I think there's a there's a deep truth that we we are just um realizations of information and particularly from my perspective of course we being a thing what what kind of information processing would be apt to describe us that could be you know articulate in terms of quantum information Theory it's belief updating so we come back to the Bayesian mechanics that you can put on top of this underlying um you know sort of information theoretic probabilistic description of the world um but I guess so reading between the lines and the twinkling your eye is the idea that he's trying to get across is that may be quite bad for Humanity is that is that the idea is it well that's what that's what he thinks and I mean he thinks you're a computationalist uh you you think that information is is the kind of primary substrate and in a way he's also worried that information is becoming the primary substrate but the big difference between you and 3D is that he thinks that humans are special he thinks that we like can't we have autonomy we choose our own actions you know we can't be replica in silicone so he thinks essentially that there's no such thing as general intelligence that there's just algorithms that perform skills and it's our Humanity which is being truncated I see right okay well I'm I would be very sympathetic to that yes I mean we are special um kinds of intelligence um and one could equip that argument with you know what is the definition of sentience what's a bright line between you know a very clever thermostat um or some machine learning artifact um or a virus and you and me and I I think there is a bright light and of course you've just said what it is it's it's the autonomy it's the agency its ability to plan and the all the existential imperatives that um that underwrite that planning and then we come back to curiosity so but if he's saying that um the fact we are here Belize the fact we are curious creatures and because we are we populate a universe that comprises creatures like us we're all curious about each other then I would certainly say yes that is a definitive aspect of us which

is not found I think um I'm just thinking Capricorn I'm sure you can find examples but um you know I don't and the kind of artificial intelligence that we currently interact with in exchange with you don't have that planning and curiosity they don't have the you don't have baked into the optimization um framing of you know what makes a viable or a good bit of intelligence you don't actually have baked in universally um this expected information came this curiosity and in that sense um I think it's probably absolutely right um and in a sense the belief sharing getting to that the right kind of belief sharing of the kind that the white paper was talking about is predicated on the notion that you will now be able to equip sentient artifacts that we make with curiosity and maybe asking well what are they going to be curious about their world what is their world it's you and me and the other artifacts so they're only going to be curious about you and me they're going to be interested in you and me um so we're talking about you know a Siri or a Google Maps and starts to ask you questions instead of you asking them questions so that I think that's that's one way of eluding his his sort of rather dystopian attitude to the the the information is King I think information is King relief updating is King that you know um belief updating is you know um of course the thing that um ensues once you act upon the world to do some smart data mining to you know to respond to some epistemic affordances um and you know the question then is your what kinds of systems do that and at the moment I would is probably just us there are other examples of some beautiful examples in um so Active Learning using machine learning you know to design your own experiments and automate the actual experiments and say drug Discovery or um molecular biology um so I mean the the there is a long history both in statistics and in machine learning of active learning um that I think does have the potential but I don't see it really being um well from what I understand in discussions I don't see that being a Bedrock of the way forward an explicit part of the design for yeah and from an age of intelligence that is um puts us sympathetically in an ecosystem of intelligence um and that's really what that white paper was trying to think yeah what would it look like to um and what would happen if your Google Maps became very curious about you particularly [Music] well there are two sort of um sort of metaphors here which are which which might sort of um ground the you know the frame um the framework or the perspective that um I'm trying to think about this this worry um one which I found very helpful um again coming from my colleagues in in industry is just trying to explain that you know the nature of shared intelligence and distributed um and the the analogy here would be the brain you know you've got really smart little elements little neurons I mean they're really smart and in fact if you get into the weeds of dendritic computation incredibly smart little

things um that are in receipt of their Sensations and they act by sort of pinging they don't know who they're pinging but they're pinging away certainly a little action potential or messages down down the axons and the wires that are emitted from from the nerve cells and you've got you know 10 billion of these things and they're all very very smart but would you call any one of them autonomous would be called would they have agency and in an elemental sense I think they do um but it's when you put them all together lots and lots of little smart things together do you get this emergent kind of intelligence that you could undeniably say um has the capacity to roll out into the future to you know generate fantasies or counterfactuals that are all conditioned upon what I'm going to do next where I now becomes this Collective so this is the kind of emergent behavior that emerges from getting lots of little smart things to talk to each other in the right kind of way so that that that's you know I think quite a helpful analogy by what is meant between uh about shared intelligence and what might you know might arise clearly a lot of these little smart brain cells would be analogous to you and me and a lot of other a lot of others would be all kinds of apps and you know uh you know giving you now the potential to see through the eyes of of any smart app that knows what you want to know and is curious about what you want to you know to find out again coming back to this notion that belief sharing I think is already there in the context of say yo uh Sat Nav you know I um share my beliefs about my preferred States would be my characteristic preferences are one part of the you know the imperative for policy selection or the good plans I share that with in terms of a destination with some shared World model or shared narrative between me and my Sat Nav um you know in this instance of a geographical World model um and then it has beliefs about the best policy it makes a little plan and then it shares its beliefs with me again so I think the you know that to the extent that um your colleague in Oxford was was uh saying we're already there I think that that's absolutely right I mean you know we already have this kind of belief sharing um I think that the move though is to make that a much more symmetric belief sharing you know I'm asking the app for its beliefs for it's I'm asking it to behave as a recommender um and um and it only knows what I sell it so it has no autonomy but if it was now in a position to actively smartly resolve its uncertainty about me the user then it's much more of a sort of you know a balanced symmetrical dianic interaction between me and the app and the agenda here is not to create paper clips or make money the agenda remember is just to resolve uncertainty to say to curiosity and to move towards a state of Greater Mutual understanding so that that that that that's a sort of the the non-dystopian view of of uh you know information sharing so um Shane Legg said that his definition of intelligence is the

ability of an agent and we're using words here like agent we'll get to those words in a minute being able to solve a variety of tasks in different environments Francois Chalet said it's efficiently creating abstractions given limited prize and experience pay Wang says it's the adaptation efficiency over finite resources so um when you look at definitions of intelligence typically they focus on principle and function which I think your one does or capability or behavior or structure now um the interesting thing about the principal one in particular is I think it's the least anthropomorphic and I think yours is the least anthropomorphic definition I've ever heard of so um and also this concept of um grounding in the physicality of information rather than reality itself and whether they are the same thing of course is a philosophical discussion that maybe we'll park for another time um but yeah how would you contrast your definition of intelligence from these other ones um I think you've already done that you you've just said it's you know it's it's a a minimal um essentially a physics-based definition of intelligence which requires you know a move or indeed a complete commitment to stay in the space of information and information geometries and belief updating um so um having said that I think all of those definitions touch upon some essential aspects of intelligence every one of them rang true you know to my ears um and I could read every one of those as being one um key foundational aspect of what would emerge if any self-organizing system managed to supervision and coexist with other self-organizing systems um from the point of view of you know of the free energy principle um the sort of first principle approach so there are no axioms or no assumptions the question is not um you know what is quintessentially anthropomorphic um the question is what emergent properties of certain kinds of self-organizing systems would qualify as having that that bimetic and then ultimately anthropomorphic aspects to them you don't even have to go to anthropomorphism I mean if you um we're talking as you have been to Mike Levin and his friends you know they would talk about basal cognition they would talk about you know uh just a multicellular organism is a beautiful construction of uh that rests upon autopriest self-assembly of individual cells but also cells of cells and you know how does a surface cell and epithelium know that it's on the surface and how does that individuate the internal cells of an organelle or an organ and from the rest of the environment so the you know the kind of intelligence that has this anthropomorphic feel I think people are are also seeing in biotic self-organization that would be a long way away from the kind of psychological psychology intelligence that we're talking about and yet it rests upon exactly the same kind of mechanics and you know for me that would be the basic mechanics that come from the uh the Dynamics of systems that are self-

organizing open systems that are self-organizing you know you talked about sort of adaptation um has been one aspect one common theme in in the um the sequence of definitions you gave I think does speak to this bright line between uh basal cognition and biotic self-organization biological intelligence of the kind you can read in many many different ways your your DNA for example your genotype is an intelligent information uh accumulating device on the point of view of evolution you know it it stores all the information about what it would require to build a phenotype that's fit for purpose in this particular environment so you know that's a kind of belief updating that's a kind of intelligence but it doesn't have what we were talking about before which is this capacity to plan Evolution doesn't plan you could also argue that the the World Wide Web doesn't plan um Google Maps You could argue it doesn't plan to a certain extent because it's certainly so what's the difference the difference is I think implicit in at least the first two of your of your definitions which is this notion of counterfactual Futures this notion of um you know imagining a future or um putting it another way having a world model or a generative model um that explains things that are not tied to the moment so if you're a physicist what you're talking about is now a probabilistic mechanics a Bayesian mechanics or possibly just you know an information theoretic mechanics based upon paths through time so we're talking about things like the pathological formulation and you know but crucially we're talking about trajectories that don't that cannot be localized to this point in time that necessarily entail the future and indeed the past so I've you know that notion of freeing yourself in a you know the account of the name of the philosopher now I can remember them I just can't pronounce it so I'm gonna pretend I can't remember well there are people who there are um there are philosophical schools that emphasizes temporality aspect now you know and if you just look at physics uh look at contributions of Richard Feynman for example yeah it's all about the pathological formulation um uh and um so I think as soon as you talk about um the elements to which the information geometry is an intelligence and autonomy um all of these things could apply are not States they are trajectories dynamics narratives paths um that have this um this sort of future-pointing aspect um then being able to select among different futures becomes an emergent property of this kind of sense making this kind of autopoiesis under red as a Bayesian mechanics of self-organization and just thinking about your definitions they all have that aspect of choosing among different futures or considering or having abstractions um uh you know about what might happen if if this so for me that that is one way of expressing curiosity because to be curious you have to imagine well what would happen if I did that and what would I know if I did that but you'll have to imagine it before it's

actually happened which you know is for me the big bright line between um you know between the anthropomorphic kind of intelligence and the intelligence you find in the thermostat or in a you know A variation autoencoder yeah and later on we will decompose the different aspects of cognition and I think as well as thinking it can be knowing and acting as well and we'll we'll talk to that but you said words you know action agent per set goal plan behavior and I guess and I I posed this to Levin as well it feels like these are terms that we understand because we have cognitive priors like a gentiveness and so on these are things that that we understand that might actually only be a lens into something far more complicated and just to touch on the um information traversal points over a geometry that's very interesting and I'm no expert but I think the medial temporal lobe deals a lot with spatial contiguity and we have grid and place sales etc etc so it you know at a macroscopic level in our brain it's a first-class citizen but there's also this hierarchy isn't there um when does it start happening you know do single cell organisms plan into the future and to the previous point is planning necessarily A reductive lens of if you feel like you brought up loads of interesting things there you caught me by the reductive lens that's a lovely phrase what does that what does that mean well um I I didn't I didn't mean it in a pejorative sense but when when we use words like you know like we we say intelligence must do X Y and Z in this plan it must reason it must act and we have this cybernetic Loop and so on and I I have a theory that this is just the way we understand things I see and and in a sense it's a lens on to a much more complicated thing and and the reason this is interesting is the reason why we have people like John settle who says that the um the impenetrable Realm of the subjective experiences Beyond function Dynamics and behavior it's a little bit extra and and even and morality is another example we'll talk about that later that feels like it's something which is a little bit extra so there's always this question of to what extent is intelligent behavior deducible from the models and the words that we use yes I think that's a fundamental Point uh you said um okay now I understand what a reductive lens is I like that I like um I think for many reasons of course because um well first of all um it certainly is not uh you I don't think it could ever be using a pejorative sense um but it does speak to um two fundamental themes which is the the way that we do make sense of our world through core screening through reducing uh through having um intuitive models of the way that the world Works um that ultimately could be seen as language words could be I think could also be seen at regardless physics and maths as well to be quite honest um you know these at least the more I read about about you know modern mathematicians and physicists talking about that you know their skills and and the the legacies that they that they enjoy the

more I realize it's all changing all the time it's just another kind of reductionism um but language particularly but it's a right kind of reductionism and I guess what you're saying is that you know um we have this um way of summarizing classifying certain kinds of behavior which may not truly reflect the underlying complexity the beauty of what's going on underneath that at that at that point I ever go the other way I would actually say that do not properly reflect the underlying Simplicity of what's going on to be quite honest uh you know and this comes back to you know the um unashamed use of phrases like sort of existential imperatives and self-evidencing yeah we're just here we're just we just have characteristic sets we're just realizations of some glorious um launch van equations and all these stories about sort of uh belief updating and um sentience and intelligence and just reductive stories that make sense of you know what we you know what um what we must um or can if indeed to a certain extent the free energy principle itself I think is a reductive story of that kind uh yeah when I say if you look at things through the lens of Bayesian mechanics as if I think the 300 principle is another example of of this kind of reductive thing it's looking at something which is inherently much more simple than the lens through which you're looking at which is the basic mechanics and the free energy principle so I think that's absolutely right a really interesting idea at another level I think it speaks to some key issues you know I mean you're um you you're often um then confronted with you know okay I'm talking with you about agency and agents and me and you uh so what license is that what aspects of my implicit World model generative model endow me with the sense of me and you and indeed me as agent and you know what would that look like if I Stripped Away um different levels of meta awareness or meta cognition if you're a psychologist and just had a minimal selfhood um you know is just be having plans um sufficient to call me an agent even if I don't know if I'm an agent if I have plans who else is going to act you know enact those plans I mean I would love to go there slightly later but there's so much we can say about um agency and the boundaries and also the causal pressures between agents and also whether you can think of boundaries as being Observer relative but I just really wanted to go to the universal um algorithm thing that you spoke about before because I think it's delicious so um I think it's fair to call you a Universalist and that and that that's there are quite a few universalists actually these are people who think there's a simple underlying principle and this is in contrast to what we were just talking about which is that the the reality is more complicated than we'll ever understand and we have a truncated cognitive Horizon and we as Chomsky says we just have kind of um simple Primitives built into us that help us frame and and understand the kind of abstraction

space within a certain code but um I was reading Professor Christopher Summerfield's book at Oxford I'd interviewed him last week he said in his book um could it be that the success of mammalian brains is not due to any careful crafting into a mosaic of different functional subsystems but instead is merely due to size we know there's a powerful relationship between the sheer number of neurons and the complexity of behavior he went on he said researchers and neuroscientists alike such as Carl Friston and uh Jeff Hawkins and even Andrew have flirted with the idea that there might be a single algorithm which underpins intelligence with the brain acting like a massive TPU repeating instructions Ad nauseam to generate complex Behavior so it's a fascinating idea is that a fair summation yes what's a TPU in this oh well a tensor Pro programming unit it's a very powerful um computer right yeah I've learned another new acronym my world is full of new acronyms right okay um yeah um so what what his argument there is it's something to do with um scale and size is is that what not not only that I mean we'll we'll get to there's a guy called Rich Sutton and he he had this um you know bitter lesson essay and it's a warning against hand crafting um structures architectures because at bottlenecks it doesn't scale so this Universalist idea is that you know maybe and and Jeff Hawkins says the same thing he's got this thousand brains theory of intelligence yes and the idea is that there's a very simple underlying algorithm or principle and you just replicate it you scale it up or out and that produces emergent intelligence right yes well okay then I am a Universalist yes um uh so you know but both of those the way you described it do speak to some um I think very pressing issues about um structures and structural learning um uh that you know you could either um read from the point of view of machine learning and sort of graph learning what's the rights you know how many layers does this particular deep learning architecture need or what kind of factorization are I going to put in play or if you uh um you know um radical con a radical constructivist this you know that's where you know I've often heard people like Josh Tenenbaum for example think about sort of structural learning and um from the point of view of the universalists have now I've learned that new word now that's good so as from as a Universalist um then you are certainly looking for uh the one principle that is redeployed at successive scales um and um that should be a sufficient explanation for those things that show um emergent behaviors at particular scales so I think that you know that that is absolutely true and again you can read this from the point of view of a mathematician from the point of view of um the renormalization group and what does that mean well it just means that you know if I take lots and lots of little things um and I start coarse grading them in a particular way um then if I want to describe the behavior of all the elements at one scale of

organization say molecules or cells or people then if I can write then down their Dynamics in terms of say A lagrangian you know some way of summarizing their dynamics that um and all the things that um a company or ensue from this don't uh that Dynamics um then if I do my core screening um and then look at the collective the average Behavior obviously lots of cells a place cell or your entire medial temporal or entire person um at a more macroscopic level then I should be able to recapitulate the same functional form of the Dynamics and or the lagrangian um so from that point of view um you have a a particular kind of universalism that is actually scale free because you get the same principle emerging at every level and that's that is basically um one um one way of looking at the deployment of the free energy principle is asking what would it look like when deployed at different scales so you can deploy it at the level of dendritic self-organization you can deploy it at the level um of um you know Neuroscience you could deploy it at the level of morphogenesis and cellular pattern formation we've done that with Mike Levin the idea being that the this the same Universal principle works at every every level then the interesting game comes between the coupling between the levels how does one level constrain an inform or contextualize the level below and vice versa you know and this I think is a really important sort of um issue which is probably well rehearsed in many different disciplines ranging from say Evolution so Evolution as a free energy minimizing process where free energy is literally the um the mark the negative abound on the negative um log marginal likelihood um I namely the likelihood of finding me this phenotype here with sampling random from a population um how does that scale of a free energy Self auto poetic process you know natural selection basic model selection if you're a statistician how does that provide constraints on the exactly the same principles um of active inference and learning in developmental time for any given phenotype and then the that would be the top down conversation the bottom-up conversation from one scale to the next scale would be you know how does my behavior my experience experience dependent plasticity mind evidence accumulation all my good pays in decisions how does that now mean I contribute to the gene pool at the uh at The evolutionary level yes so that yeah that that would be what we're reading it the other way of reading it of course is just if you're designing um a TPU or deploying a TPU um you've got you've got message passing on some graph what is a graph well it only has interesting structure in virtue of the um the sparsity or the connections that are not there otherwise it's a full graph and it's not very useful for anybody speaking to Chris's Christopher's uh uh you know you know two two big too many neurons um you can't you you necessary to have to have a sparsity to fit all those neurons into I would put it the

other way around though I I would say that anything that is adaptive and has this size uh you know what properties must it possess and I would I would suggest that it has to comply with the principles of um self-evidencing um where evidence now is the marginal look the marginal likelihood that can always be read as accuracy minus complexity so as if it exists and it's big it's got to be minimally complex what does that mean it's going to have the smallest degrees of freedom the minimum number of Connections in so you should be able to predict the sparsity from the first principles at every scale so that sparsity defines the nature of a graph and indeed if you're talking about anything that's deep in a hierarchical sense all you're saying is there's a particular kind of graph that I have in mind and it it you know has a certain sparsity structure but crucially it's a sparsity structure that allows me to call it a hierarchy it means that there are no connections that transcend unlike a sort of you know a U-shaped uh what is it no it's still hierarchical to anyway sorry I'm getting a bit distracted like the choice so I mean I think that's sort of um the notion of um coupling between different hierarchical scales is absolutely crucial from many different perspectives renormalization group Evolution um you know getting the right graphical architecture on your message passing scheme and computer design so yes I wanted to bring that up but I discussed exactly the same thing with lemons so morphogenesis and the rungs of the emergence ladder and the causal pressures between those rungs and you know philosophers like George Ellis said that you only have causation between the levels and Douglas Hofstadter and go to *Leisure bark* and the strange loot is that there's a very complex Panner play of causal pressures between the scales and like that I gave the example to Michael my mind is an emergent phenomenon and I command in my hand to move and he said that at different scales you get different amounts of work and actually I think if you get into integrated information Theory it's kind of talking to that a little bit and and I think you think of Consciousness as having a lot of information processing going on because it's at the top of the stack to some extent but do you have any intuition on how that information is kind of partitioned between the scales and how those causal pressures work between the scale yes I do wonderful you're very well read aren't you uh so it's nice that you you mentioned George Alexander I use the word top-down causation exactly the spirit that he writes about it it's wonderful I have literally had him in mind but yeah so I was hoping if he ever hears it he'll he'll know that I was talking about him so he's exactly that and it always makes me a bit queasy when I use the word emergency which I think since some people say there's no top-down causation in emergentism but I I don't fully understand the philosophy of it but it just don't acknowledge that sorry yes I've distracted myself from your really important question

which oh yeah see um so the the different scales um of a Bayesian mechanic self-organization viewed through the lens of basic mechanics um I think um what we've just been talking about and I would imagine with um with Mike as well a lot of focus of uh here in that kind of work um is um I hesitate to use it but I will use it spatial scales you know how do elements how do single cells assemble into multicelli where why what we are how on Earth can you um envisage the emergence of multicellular organisms those Mike's done some beautiful theoretical work you know several years ago now just just think about it to be a skin cell to be an epithelium means you have to sacrifice the ability to reproduce so it's if you like completely paradoxical from the point of view of natural selection you have to sacrifice yourself with a greater good so there are you know there's some wonderful questions about uh about um cells of cells and of cells and cells as you build up to different levels of spatial scale but I think your question will be better addressed from the point of view of temporal scales um and again you come back to this um universalism um now I'm getting fluent using that word um where you've got the same principle playing out yeah exactly the same principle exactly the same mechanics the same lagrangian um playing out at different scales that um where each scale contextualizes and has this circular causality at the bottom map and top down um aspects to it in play so my favorite example of this is just to um look at a succession of belief updating processes from the very very fast which would be from the point of view of um your sentient machines it would be inference inferring states of the world as they are in the moment so State estimation Bayesian filtering um everything that um you know speaks to some kind of situational Awareness on the basis of some smart and hopefully smartly sampled data and then we move to the next level I'm going to skip attention and precision but there is an intermediate level which usually um in in neurosciences has a time scale of you know your hundreds to milliseconds to seconds but I'm gonna I'm gonna jump straight to learning so what's learning well it's just slow inference it's just basically um slow belief updating um where the states that matter now are equipped with another kind of label which we call parameters or weights in machine learning in a neural network um but they're just random variables that are brought to the table to explain uh or part of our world monologi model but they're special um kinds of parameters because they change very very slowly um and then you move well okay so those are two levels what about turtles all the way down and Turtles all the way up okay what's the next level well the next level is now the structure so now for any given graph for example um that is equipped with edges and those edges will have to have some um slowly very varying parameters um that describe the you know um the nature of the message passing on those

edges um there will be um there will be a you are conditioning up on a particular structure and you know is there a connection there is it a hierarchical is it heterochical is it you know are you net is it um is it a Transformer it was a convolutional neural network you've seen this wonderful evolution of structures in machine learning over the past few decades as people try out different structures and you know somewhere for one kind of application of the work for others but um from the point of view of your question what we are seeing is a kind of structure learning that's playing out over years so this but it's exactly the same principle it's this free energy minimization but just in this instance the free energy now is a pathological it's just the average over a long period of time which is the um which is the exactly the quantity that people doing um structure learning or Bayesian model selection use when adjudicating between different graphs usually um of complex system models for example um it you know and you could argue that now um over um several tens of years or hundreds of years that you're exactly the same maths could be leveraged to provide a formal description of natural selection as Bayesian model selection exactly the same things you're doing now when you're selecting whether to speak or whether not to speak or trying to infer selecting the right hypotheses about you know the narrative that you you have in your head that makes sense of what I'm saying um exactly the same maths and mechanics is unfolding over the Millennia um you know in terms of in Terms in terms of evolution so I think that's a nice example of the separation of temporal scales but the conservation of exactly the same principles that have this information geometry and implicitly intelligence in of a basal sort I use the word basal because that's what Mike Levin and Chris fields and Jim Clay's really like using is it's this notion that um basal cognition and basal intelligence transcends physics psychology and biology um it's all the same thing this is a line from Chris Fields there's a friend of uh um and I think that's great A Great Notion um and I think one can do that very gracefully by being a Universalist and just by finding the right principle the right sort of reading of Dynamics and you know that reading you know for me is the information geometry that um that that supports the belief updating I was going to do Markov blankets later but it feels relevant now and maybe we should bring a bit of continuous versus discreting so a few things came to my mind when you were talking about that first of all we think of emergence over time and self-organization over space and I guess it just occurred to me that are we only interested in time and space when we talk about this kind of structural learning and then um with these Markov boundaries I had only thought of them in at one point in time but you could actually think of a kind of three-dimensional markup boundary um over over time as well now just to remind our audience um

blanket States facilitate the interaction between the internal and the external conditional independencies the external states are independent from the internal States as long as we know the intermediate blanket States now um to get to the uh to a core issue that we've been modeling complex systems you know like where do you draw these boundaries and is it boundaries all the way down yes so I think you're absolutely right this is the perfect time to bring up the sort of uh I hadn't really thought about the notion of boundaries in time and sort of uh um that's intriguing so but you distracted me I'll think about that I'll think about that after our conversation um but no certainly so um there's certainly a lot of current interest in um taking the notion of Markov blankets which is foundational in this sort of reductionist lens of the free energy Principle as a Bayesian mechanics um you know the one could could summarize but the basic mechanics of the free energy principle is simply just another kind of uh quantum mechanics or statistical mechanics that inherits just from this partition that is the Markov blanket that separates the inside from the outside yes um now um and of course there are lots of next issues about one how do you identify those Markov blankets and how long do they endure for um so there's lots of interest in that at the moment but one very simple approach to um the question how long do they endure for is to say well that's not the right question because we've just talked about separation of temporal scales so you have to say at what temporal scale well you uh operationally Define the Markov blanket as the 10 the time over which it exists uh and what would that look like then when you're suddenly now think um this situating that Temple scale within the context of a larger temporal scale so what you now have is a picture where big mark off blankets blankets and blankets things that define say you were you and me or cultures or Nations States or institutions that um outlive say species these big ones uh last for a long time but they contextualize and provide constraints on the mark of blankets at this at a smaller scale that last for very uh for for much um much shorter periods of time and so on all the way down so that at some level say at the the molecular level from your perspective these Markov blankets may only exist for nanoseconds or your milliseconds um but from the perspective of the molecule thank you yeah this this is a lifetime and it's well Happy complying with or can be understood as um you know doing its own basal intelligence doing its own basal belief updating just automatically for its lifetime which may only be a few hundred um you're making sense of its world or inter being interpreted as having the sort of um sort of you know biotic intelligence and self-organization just because it exists for that period of time so then again this interesting question you know how does one time frame contextualize the other starts to bite and you know you start to now think um from the

point of view of a slower time scale what would um a succession of Markov blankets uh look like and what one immediately um encounters is the notion of a of a Wandering itinerant Markov blanket yes we spoke about vagueness on on the first one so we'll we'll park that the wandering sets is very very interesting but um I there were two things that I that I think I wanted to understand um you do think that there's a hierarchy of blankets but I'm interested in exploring this idea of whether the blankets could be Observer relative to use the word could be understood as a mark of blank there and that brings up two things to the floor of my mind first of all the extent to which they are a lens versus describing something in reality I see and if they do describe something in reality you spoke about this symmetric causal pressures which we can we can speak to as well but but on this understood thing it's very similar to um Wittgenstein said the meaning of a word is in its use and that meaning in language is is embedded in pragmatic actions and the language game and so on and and then like the the further thought occurs well maybe the the understanding of a boundary or a you know Markov boundary blanket could be um understood in the context of one's perspective right that's great question and um and I I may be informed by reading some of the philosophical literature and I should remind you of course I am not a philosopher so um so that's another one I say will be naive um you need to speak to philosophers about this but I would I would say the Markov blanket um is is something which is metaphysical um you know it it is defined by a particular spasticity structure when formulating any state space in terms of Dynamics and specifically you know a laundroman equation so one talks about the free energy Principle as a first principle account um but it does actually commit to something it commits to the notion that there are states and that those states have a separation of temporal scales that disambiguates or separates systemic States from random fluctuations so the that's all it does there are no more assumptions and then everything else for us from that under the understanding that fourth sufficiently large systems the probability of there not being a Markov blanket is um zero um so and this comes back to the sort of um this uh sparse coupling uh conjecture um that Christopher I think was alluding to um or at least we um we unpacked in terms of this you know that if a system becomes exists of and is sufficiently Big it will be sparse and once it spars there um with probability one will be Markov blankets so at no point now have we introduced the notion of observers we haven't at this point even introduced the notion of the free energy principle we're just saying um there will be one way of carving up large dynamical systems or certainly systems can be expressed as a launch of an equation or a random dynamical system um will be ways of carving them up into one or more Markov

blankets at one or more scale yes um and then the question is well what um what would it what what would one uh how would one then describe um the Dynamics for any given Markov blanket so you at this point you are at Liberty if you are simulating a universe to choose the Markov blanket and the scale that you want to simulate if you're part of the system you're not at Liberty because if you're part of this system you have your Markov blanket so all you see are the sensory Impressions upon your Markov blanket and you may or may not have active States and it pixels back to the southern bright lines that make between agents and sense making machines that can't act sessile things so if you're asking I um are the Observer dependent perspective uh dependent aspects of this formulation using Markov blankets as a epistle logical device not if you're part of the system it says something quite profound about what you could you know what you know what and if you're being observed by something else if you believe if your world model is populated with the fantasy that there are other observers out there yes um then those observers will never ever know your internal States well what about rather than an observer um an actor or an agent and we'll talk about an activism in a little bit but even at the microscopic scale there are still affordances and you gave this beautiful example of um of a type of species where someone might kill themselves with a great good and that that shows the kind of information sharing that you have in these complex systems so similarly if I'm an agent in this system and and I act and and there are these you describe this cybernetic Loop and there are these kind of symmetrical causal pressures and so on and and and then you almost get this emergence of the markup blanket so it's changeable yes no it certainly um that's certainly true it is changeable and it's malleable and it's self-assembling uh and certainly when you have Markov blankets and Markov blankets and coalitions and mark off blankets for example or formation of in groups and out groups and the like you would expect there to be a a you know a dynamics of of the Markov blankets or who you're relating to so you know the sparsity uh structure and who I talk to and who I listen to and you know what social media uh you know I I commit to all of these things are um products of or Reflections and can all be articulated in terms of you know wandering Markov blankets now that that's absolutely true um I I thought you were you you were asking something more about um whether the Markov blanket is part of a realist metaphysical description or whether yeah I mean there is another argument that if you if you now play God and you now put your Universe in in a computer and now to simulate and and write papers about your simulated Universe where you can see the inside you know the internal States next to all states then you know I think you're in a different world um just again I repeat that

something quite profound about the um the questions of you know what is it like to be something are interesting because from the point of view of the free energy principle what is it like to be something means that someone's asking that question so they are a thing and they're asking about another thing so now they're asking questions what is it what um what is it like to um know the internal states of a Markov blanket but by definition you can never know them so there is something impenetrable about being something else so it's a question which is unanswerable it's unknowable um so if um from that point of view I think there's something uh beyond the um the sort of the epistemology of you know of Markov blankets but if you're just simulating cells or people or you know doing multi-agent simulations then obviously you could you can choose where to put your blankets and you can you you can just use them as a device to um to understand self-organization to simulate it and to predict and I think in that instance you know the you you could deploy your mark of blankets wherever you wanted to whatever any scale you that you you thought was interesting for the phenomenon of Interest that's not quite speaking to what you're into there which is a changeability of the the mark of blanket so well I mean we can get into that um because if uh and later on we'll talk about the the real world implementations of these ideas but um but it but if you if you did hard code the the Markov blankets then you would bottle neck the system essentially so ideally you you would actually have the system defined at a sufficient resolution where all of this could could emerge um itself yeah did you cover the point of whether the blankets can be um must they be spatially temporally con you know should they have a spatial temporal contiguity and the reason I'm asking that question is when you think of a blanket I like it's almost synonymizing it with an organism and maybe that's a bad thing to do and you can correct me but but if we think of um discontinuous blankets in information geometry that's an interesting idea because we kind of visualize them in 3D space yeah yeah well certainly they'll have a topology yeah um I mean I've noticed earlier on you're talking about sort of place cells and grid cells which are you know very special and beautiful constructs which speak to a certain continuity or contiguity aspect yes they suggest that these um you're coming back to uh Christopher some of um uh summer fields um book um you know one way of understanding the um the Markov blankets in the brain so you know we've been talking about Markov blankets delineating me from a world but of course um your temporal lobe has a Markov blanket as it wouldn't be a thing he would be able to call it a temporal though so you know the very hierarchy could also be described now in terms of um uh Markov blankets within Markov blankets within Markov blankets and some really interesting questions about what that means

for what what part of the brain can know about another part of the brain and what and what action becomes and constantly becomes attention yeah I just wanted to slip that in because I think that's what was in part what you were driving out with this sort of context-sensitive fluid dynamics on in exchanges between Markov blankets I think that's incredibly important even within the brain in Neuroscience this is basically the the routing of information the deployment of um selective attention or sensory attenuation attenuating ignoring some signals it is exactly that that makes us such adaptive context-sensitive information processing um machines and of course that is exactly what you'd have to worry about when designing a web with routing and context sensitive you know who do I listen to uh so you you know this is really really important um the um so the the the the the emergence of um I think you're absolute right they will have a topology and if I were using Markov blankets and you know I do and many other people do every day when um thinking about message passing as a statistician on Factor graphs or uh um which are dual to a graphical Model A probabilistic graphical model of the system you're trying to estimate or understand then you know the markup package is just the um the implicit nodes that that I you that influence me or that on the parents and children the parents of the children yeah and that has enormous implications for minimizing the complexity of the message passing um and you know and I repeat defines the hierarchy for example um so Markov blankets there will have a topology um probably best understood practically from the point of view of graph Theory as opposed to uh um information geometry but there are special kinds of connectivity that you see um such as Place fields and I'd also say if you just look at the history of machine learning there are particular um kinds of Markov blankets that characterize the very structure of certain neural networks I'm thinking of convolutional neural networks I'm thinking about you know often motivated through weight sharing and the like but you know if you look at weight sharing as just the kind of stratulations trying to minimize the complexity part of your free energy or your um negative log marginal likelihood um then what you are doing is finding the right kind of sparsity fit for explaining these kinds of data what uh what what what is the architectural principle it's it's the translational invariance of translation symmetry it's just the contiguity aspect it's a fact that you are sampling data that is generated in or from a metric space that has a well-behaved metric so I I would say that sort of you know Place cells and grid cells and many in many other are thinking about we have Chris's friend Tim Baron's work for example um finding grid like structures everywhere yes in abstract spaces uh so what that tells you is that that it's likely that we live in a world um where data are caused by samples from a metric space that has a

you know a a measure to it um not all spaces do um I think it would be hard pushed to find your the right kind of metric for language for example um and certainly if you're if you're if you're sort of building message passing schemes or um believe propagation schemes on a factor graph you don't you don't think about contiguity in a metric sense you know you don't measure the distance between one node another node is it connected or not um so I think there are some special um contiguity let's say there are some special worlds that are recapitulated that emerge in the internal architectures of intelligent machines that have this contiguity property in virtue of the fact that some aspects of the causes of the sensorion are elaborated in a metric space but not necessarily everywhere and then well I want to take you back I'd like you for you I wonder where you want to take me anyway well I'll finish them because I see uh in the brain and I also see in the direction of travel in terms of um building thinking machines um as you get deeper into these hierarchies and you go from uh fine scale to larger scale there is an interesting move between architectures that would speak to this metric aspect in space and in time to a more discretized topological non-metric representation yes I know I wanted to touch on that I mean there's a beautiful example I mean um Christopher Summerfield talks about Pierce's Triad and various ways that that we learn different types of um symbol and abstractions and so on and there is a famous experiment in Neuroscience I'm sure you're familiar with it where they showed um people pictures in a certain order and associations and then the same topological structure was was recovered I think in in the ntl and and this is because even though with spatial temporal contiguity there is a metric space but the brain learns these kind of associations and then you can essentially learn these abstract Concepts that you know Concepts that reverberate in our language and experience they get represented yes and I completely agree with your um point about the the Revolutionary idea of the um the translational local equi variants you know the the weight sharing with cnns it's a beautiful idea it's revolutionized deep learning so um just to finish off the discussion on Marco blankets I wanted to talk about part whole relationships so um for the benefit of the audience in logic and philosophy myriology is the study of parts and holes that they form whereas in set theory it's founded on the membership relation uh between a set and its elements in myriology emphasizes the relation between entities which is to say the inclusion between them so anyway when considering systems Markov blankets can Nest into hierarchies and what connection if any is there you know between that and the philosophical study of mariology and um I was also going to bring in I don't know if you've heard of hinton's glom architecture but it it was it came after capsule networks and it's you know we were just talking about these

wonderful inductive priors and deep learning and and they yeah they make the problem there are many curses of machine learning but one of them is kind of like the curse of optimization and um and there's a complexity curse as well and that's why most of these inductive priors they reduce the size of the hypothesis set if you reduce it too much you get approximation error which doesn't help you either there's curses everywhere but um but this is a really interesting prior as well these part-whole relationships gosh yes there's so many issues you bring that I'll just reiterate my favorite one which is the the the notion of um these inductive biases minimizing the complexity I think that's absolutely you know that's something which is absolutely Central um from the physicists perspective you know this um reading self-organization as self-evidencing which is a philosophically poetic way of simply scribing existence as an optimization process which you shouldn't really do it's all it's really a principal at least action but you can read it as an optimization process what are you optimizing the evidence for my world models what how can I carve up that evidence complexity and accuracy what does having a what would I then mean by optimization well I mean just providing the simplest account that maintains a degree of accuracy what do I mean by simplest minimizing the degrees of freedom that I use up in providing that accurate account how do I do that by the right sparsity structure what does that mean it just means finding the right structure and you know you could actually think of much of evolution and the trajectory of machine learning architectures as this game of finding the right structure that are just these structural priors that are apt for describing the kind of data you know so if you're you've spent your entire life doing mnist images then it's going to be the kind of structural prize that they are that that inherit from being sampled from a metric space and you'll have all the convolution I think the capsule I didn't know I didn't know about the glom stuff I always like listening to to Jeff's um um latest ideas because he's always got the right kind of intuitions and and these intuitions are part of the I think um part of our reflection of um Ashby's law of requisite varieties exploding hypothesis space about the various structures uh yeah um but certainly the the capsule stuff and the you know the um one aspect of structures of graphical models or implicit Factor graphs when it comes to implementation which I think is often neglected um is the orthogonal direction from uh orthogonal you know um in relation to hierarchical composition and that's the sort of the breadth of a model in terms of um having factors that can be separated so if you wanted to have say scene construction you wanted to have um multi-object you know the ability to track and infer and and um make sense of data generated by multiple objects what he is saying well there are multiple things out there and

these things um again from the point of view of first principle account have certain conditional independencies that is literally revealed by a lack of Connections in your world model what would that look like well if you're a physicist it would just be a mean field approximation literally factorizing um introducing conditional independencies and factorizing a massive joint distribution any one level in your hierarchical model into a number of factors that if you're a neurobiologist would now be functional specialization and modularity of a fodorian sense you know what and where in the brain would be the the conical examples so I looked at that as um as a move towards paying more careful attention to the non-convolutional aspect you know those things that that actually deny a translational symmetry tree and actually let's celebrate the conditional independencies within any one particular within one particular scale so my guess is that I have no idea so I'm talking from a point of view of complete ignorance but my guess is that um if it's an extension of capsule uh networks and it will have that aspect it'll have that separability that sort of things that can do stuff and account for attributes that are independent from other attributes or other objects that are that are conspiring to generate uh the data and certainly my world of toy prototypes generative models usually in Matlab um you know then all the heavy lifting is done from the the mapping between the levels again so all the interactions you know big red buses there's bignesses rednesses and busness these can be factorized but they all have to conspire literally through interactions and highly non-linear um operators um and then generating what I would see if there was a big red bus um so you know that puts a lot of pressure on the the likelihood tenses or the mapping from you know you'll say a sense an input layer to the the first hidden layer for example um and then the sort of leads you into all sorts of interesting issues about you know how do you accommodate that non-linearity and do you uh again as looking at the evolution of machine learning architectures as an evolutionary process you know a relative or you know 108 whatever um you know that is another aspect of this sort of structure learning I think will be very much finessed I think if we just move to sort of quantum operators and and just go to discrete State spaces and in the spirit of quantum blue gravity I think sort that out and then we can worry about what particular nonlinearities uh anyway so I wondered away what was it what was your question my point well no this is a wonderful break point so um before we hit record we were talking a little bit about chat gpt and just before we go there because that's a fun discussion I think one of the things that really distinguishes your your line of thinking obviously there's the uncertainty quantification so on but some also there's this idea of an activism and I wanted to just do a Whistle Stop tour of that as a contrast to these more monolithic approaches to AI like like chat

GPT so um an activism contrasts with representationalism which is this idea that you know I I know everything about the world inside the model and um you said that you can't just think of the brain as some behaviorist thing it's a dance of dialogue you act in the environment and the environment acts on you in a cybernetic Loop now you also gave the I'm quoting like an interview that you did previously you gave the example of radical inactivism that you can dispense of representationalism entirely and you gave this beautiful example of this walking robot that kind of fell down a hill and it did so so gracefully it looked like it was walking it was all in the body um you know if the body is sufficiently tuned to the environment you don't even need cognition and of course we'll talk about a decomposition of cognition in a minute um and and you bring in this idea of circular causality uh so we're causally embedded in the world bidirectionally essentially now when we decompose cognition um I think of things like thinking and feeling and knowing and acting and and the environment and and so on and I I think this is one of the key things that distinguishes your line of thought so could you give us a bit of a Whistle Stop tour of an activism yeah so um an activism is at the heart of um the circuit causality that that um follows from the very existence of a um and a Markov blanket is the thing that would be it's certainly in my world necessary for the existence of something that is demarcated or individuated from something else so just having a separation between thing and nothing or not a thing um having uh that separation requires you now to think about the two-way traffic um the bi-directional traffic and of course now you've got um two directions of travel that can be thought of as um the agent if you like sensing the environment on the one hand on the other hand the agent acting upon the environment or vice versa you've got now a circular causality so coming back to sort of the other the the the the the notion of a perception action cycle and this notion of a dancing and a dadic exchange which is bi-directional two-way traffic uh between the two so when you apply the free energy principle in practice to um emulate or simulate sentient behavior and behavior that is predicated on sense making um then you are necessarily simulating action perception Cycles so you're necessarily inactivist and that's called active inference so sometimes just for fun I put an En in front of it called inactive inference uh I have to say the active inference was really a nod to Active Learning it was just it's the same idea but um but cast in terms of fast belief updating as opposed to slow endless accumulation in terms of learning contingencies um so that use of inactive is at the heart of applications of the free energy principle and obviously has been at the heart of I think all right-minded formulations of behavior and self-organization since Plato probably but certainly you know things like perceptual control theory cybernetics and you

know all of that good stuff everybody at some point active sensing well you know we'll we'll um have to commit to this um active aspect um even to a certain extent semiotics I think well now perhaps we shouldn't gather um but the so that would be one way of um sort of celebrating and foregrounding the role of action what would that look like from the point of view of uh machine learning and um computer science well it would look like basically smart data mining um and it would change the nature of the game from um Big Data making sense of big data so having everything on the inside having access to the entire world um and then making sense of that and you know you may ask what does that mean by making sense of it what's only doing something with it like generative AI um versus um the um the complementary approach which is much more in line with this sort of complexity minimization and the imperatives for the sustainable self-organizing self-assembling systems versus smart data um so the job the action now is it's basically what moves do I make on the world to get the right kind of data that will serve my imperatives what are my imperatives to maximize the evidence for my model in the world and if I can do that by definition my um I will be there or put another way around if I you know if I exist then that is what it looks like I am doing so you know you talk about the or you mentioned this notion sort of Monolithic big systems um versus small agile intelligent little agents and go and get the smart data that they need or that they think that you need that is exactly um the sort of picture that underwrites this notion of distributed cognition and a different kind of network and a different way of relating to intelligent artifacts and information services and um and apps uh where it's lots of really small smart things who are actively getting the right kind of data that they need to resolve uncertainty about the context in which they find themselves a game that you know the complexity minimization gets in and everything is really I'm sure we've talked about this before but I can't resist just um mentioning it again from the point of view of sustainability and climate change you know the direction of travel of these large and say large language models for example is it's so wrong uh wrong from the point of view of the ideology of climate change but it also wrong from the point of view of lamb Downs principle should Escape equality when read as a thermodynamic corollary of self-organization and non-equilibria you know you've got to minimize the complexity minimize um all of the internal machinations so that you you just need the minimal amount of data expertly handled every little agent is a good scientists designing the right experiments and get the smart data that resolves it's uncertain about what it doesn't know uh and then job done if you if you can do it like that so that would be one um if you like sort of answer to your question you know um the implications and the

importance of inactivism used really just as a euphemism for an agent that can gather its own data the radical activism is I think a more of a philosophical thing and it's more of a fun argument and I don't know any radical in activists so I I you know I I don't I don't have um I don't have the right sensibilities uh it's interesting to really answer this question but it's from what I understand um that you know they have um taking it a little bit too far and a denying representationalism um uh to the extent that you know that you can get a kind of um sense making that doesn't actually involve any internal Dynamics um and um you know I'm sure that's true and I'm sure that you're going to sell um uh chat sheet GTP to me that's what one example of this mindless kind of uh sort of you know enacted uh uh sense making uh which is the opposite I think um I would call chat gbt extreme representationalism all right yes we'll resist the urge that we'll go there immediately after this I I promise but um so clearly we we think that intelligence does necessitate embodiment I think that's clear I want to just explore this Continuum between um and activism and representationalism this is this is really interesting so um this comes down to grounding to a certain extent so cognition must be grounded in different domains and in the physical world possibly in language in acting in affordances in knowledge as well and so this is this is a view that that it that it is grounded let's say in in affordances but if these agents these organisms ground their cognition in affordances then to what extent could you say they are learning a world model if that model is with the lens of an affordance I think I think even sort of um reconcile um very rich representationalism um in the service of inactivism Simply by noting that um in the circular exchange you have to deploy the right kinds of actions and that's going to require um coming back to what we're talking about before in terms of planning being able to build the right counter factuals you know and to do to do that in an expert way in an intelligent way you need to be able to represent the causal contingencies and states of Affairs in the world at this time upon which you're predicating your next action so I I'm now thinking this looking at your your expressions and and I I now realize that you've that you were trying to sell a dialectic between an activism representationalism for me um they are the same thing uh you to be a good in activist you have to have the right representations if you want to well may I say good I mean um good at a particular scale the big things that survive so viruses don't need to do much planning um but things like you and me do need to do kind of planning so the As you move up those scales you have to look further into the future so that we certainly need representations which is what I have which um which is why I'll smile always smile when I think about radical and activism so if there's no space in radical and activism for being able to plan to

imagine scenarios to have narratives that play out on the inside before committing and selecting the Right Way Forward well it's not an app description um certainly cognition okay it's so interesting because the the question to me is we just spoke about this idea of planning potentially through an information geometry and many of the abstractions that humans learn are not grounded in the physical world at all and that's very interesting and we can get to how those abstractions are learned but I guess what I'm saying is let's say one of these agents in this multi-agent system it's traversing this topology what grounds the states in that topology well I think that that would be um ultimately it will be the transactions um you know with with the world that underwrite um at the lowest hierarchy level all the generative models of of this hierarchical sorts of the abstractions are just the coarse graining simplifications that you know the the um the products of literally looking at at the lower levels of your hierarchy through a reductionist lens in the right kind of way that enable you to carve up the world in terms of these um in terms of these abstractions which may live in a non-metric space um uh they may still have this ordinal structure you referring to before um and then what that would mean in the context of um lots of similar artifacts who had done the right kind of core screening would be now the opportunity for uh direct belief sharing a brain to brain communication uh speaking to a friend of mine even as they talk talking about the distinction between communicating through the through sensory um exchanges such as language but what we're actually doing is basically directly exchanging beliefs um of course you could do that directly in silico you could actually have messages about what this agent believes at this abstract level of um of representation very high in a you know a hierarchical graphical model it could pass its sufficient statistics to another partner somewhere else in the world Elsewhere on that on that graph so you do have the opportunity now for direct belief sharing um and all sorts of interesting questions about you know how you'd engineer that you you know you can't coming back to this fact you have to have spasticity in the game you have to decide where to to send and receive your beliefs from then we have this sort of attention and routing problem and then you have do you have peer-to-peer communication do you go through a server can you write that down in terms of um Quantum information Theory as a holographic screen and who's you know having all sorts of really interesting things but what they speak to is that you now you're sharing beliefs literally basically it's probability distributions as encoded by sufficient statistics that can be passed um as messages uh on a on a factor graph um you're now talking about the ability for proper communication of the kind that has evolved in terms of cultural Niche construction and evolutionary psychology that speaks to cultural Niche construction that we

enjoy in terms of you know the the words that we use and the exchanges that we use yes yes indeed so that there are two things to bring in so knowledge and language and you did invoke Andy Clark in another interview which is the extended theory of mine so it's um one of the five e's in in cognitive science and the the communication substrate could of course it's it's distributed so the two agents could be referring on missing information that actually exists perhaps in another agent or another another knowledge repository and then we can also talk to exactly what the role of language is and how it's compressed how it represents abstract Concepts and so on um so I'm very interested in knowledge and language I'm not sure if you could bring those in right um I'm not sure I can do so expertly but certainly a part of learning new things since my foray into into industry is this notion of knowledge graphs so if you know if you read a uh a probabilistic graphical model as um at least its structure as being a Knowledge Graph then embodied um in the structure and presumably the parameters of the connections that constitute that graphical model that would be knowledge so for me um in a very non-mysterious and possibly too simple-minded way knowledge is just the product of um belief updating of a slow kind which is learning so I know contingencies um in the sense that I have a suitably configured and optimized structure and parametrically weighted um generative model that can be written down as a knowledge graph or a graphical um upon which I do my message passing to do my inference about the particular context sensitive in the moment kind of thing so I would put knowledge um basically as an attribute that is implicit in a um a graphical a description of an implicit generative model or World model that is actively that is used actively to do the data mining and to do the do this inference and and the sense making and then the language part of it um would just be that highest level most coarse grained summary that is conserved over multiple agents so just by definition um if I want to do belief sharing um I have to um I have to have a a shared generative model or a commitment to the same narrative so that the meaning of what I'm emitting is received in the right Spirit or the right frame of reference by you and indeed um there's some lovely rhetoric from Quantum information theory of the kind of Chris phrenals and Jim claysbrook have been pursuing where you literally have to think about the geratin model as a Quantum frame of reference and we have to share that in order to communicate so in this instance the um the Markov blanket cease ceases to be just a set of states and see and and adopts the role of a holographic screen and action now is writing to that screen and sensation is now reading from that screen and there are two agents on either side so you know whatever is written is an action but can be read by something else and then you're looking at the entanglement which is the synchrony of mutual

understanding that you know we would aspire to uh through through uh through through communication so that that that um but that only works if if the messages that are written to the screen or written to the Markov blanket um have the same kind of interpretation so it speaks again to this um the fact that you have to have a good model of the world and the world has um and that really means that there's a kind of entanglement or generalized synchrony from the point of view it didn't have consistency between the two sides of your screen or the two sides of your Markov blanket or your server which have a you know the right kind of isomorphism so we're talking about you're basically a shared narrative that underwrites any exchange of signals and again what you know just thinking of of this from the point of view um of generalized synchrony what we are talking about is just the characteristic uh or the emergence of characteristic behaviors in any sparsely coupled set or Ensemble of things that possess Markov blankets or um have a separability um what will they ultimately do they'll ultimately find a a synchronization manifold they'll find a synchronization simply because um this is the um the most likely state of being and the most likely stated being is that which maximizes the marginal likelihood which minimizes the free energy which just means that they Now understand and can infer each other so you know it's just another expression of this existential imperative to resolve uncertainty you have good models of my world and my world also needs to have good models of me and if my world is you you have to have a good model of me and I have to have a good model of you ultimately what that means is we're going to converge on the same model um yeah I mean what interests me there is that unlike many other researchers you're com conflating understanding you know knowing and intelligence itself I get I guess so yeah um so good question um if intelligence is a process of belief updating then yes it's just intelligent intelligence and learning I just learning knowledge and intelligently inferring states of Affairs are just the same process at different time scales and they both depend upon each other so I have to have the right Neil network of the right weights and the right learning to make sense of the data in the moment and if you think of it from the point of view of uh weight learning um either through propagation of errors or through experience dependent or Spike timing plasticity in the brain you have to have the right inference in order to do the learning so there's again this circular causality between between the two scales so knowledge requires the right inference or state estimation and state estimation requires the right knowledge to make sense of of the data that's being assimilated yes although I suppose philosophically we could break it down because you know knowledge might be the sort of the information acquisition because knowledge exists it's quite an interesting illustrial point

actually that knowledge is on Wikipedia and it's crystallized knowledge it exists as a thing but but the ability to to acquire knowledge without surprise is is is your your form of intelligence but I wanted to talk about chat GPT just quickly and it's very interesting because Bing have just released a new version of their search engine which integrates chat GPT and I've I've been on a bit of a journey when they released gpt3 in the it was November 2020 I got access to it I thought it was garbage it was just generating a load of rubbish basically but there were people out there who were True Believers and they said Tim you're not seeing it I you know I I I've seen it and they would show me these ridiculous examples and I just thought no you're just Fooled by Randomness and um and then DaVinci 2 came out about a year ago and then that that transgressed the anthropomorphic Fooled by Randomness threshold so I I started to be a bit of a Believer I knew it didn't understand anything but it started to get very useful when I was using it for coding and generating emails and so on um much to my loss actually because recently I've been checking code into the to the repo and my colleagues have been saying to me Tim it looks like you used GPT to generate that why is it full of holes and you have to hold your hand up in a minute oh yes sir I'm sorry I just I just checked in some code that I clearly didn't understand and uh you know didn't actually save me any time so sometimes you can see problems with its generation but it's so plausible that most of the time they're hidden and unfortunately if you want to verify that knowledge you might as well have just not bothered using GPT in the first place because you could have just gone so a little recap it's using a self-attention decoder Transformer and that's a neural network architecture that introduces uh permutation invariance to Tuple permutation invariance which turns out to be extremely useful for language and then there was this discovery of what's called in context learning which is where rather than just getting it to because it's a generative model it just generates token token you you insert a prompt and then continue to generate from that prompt and people discovered you could ask it questions it had this emergent reasoning capability in Big Air quotes and then more recently people have done what's called preference fine tuning which is that you do some additional supervision on the top with human reference examples and that aligns it to humans and makes it give slightly more politically correct or you know more sensible answers so and now it's been integrated into GPT and that does this retrieval augmented generation which means rather than just being a snapshot in time it will also go out go out to Bing get some relevant search results incorporate that into the prompt and then generate from there and there was this incident a couple of days ago where Bing had this successful launch to much fanfare and then people looked at the results it was generating

including fight you know one of the the things was give me um a comparison between the financial results of Lululemon and some other company and it was just hallucinating the the results it gave weren't even in the document and the product managers at Microsoft didn't even themselves bother to check the truthfulness of this generation so God help anyone else using Bing um so what's your take up I love that story thank you um as we were talking about before I I've heard so much about Chat GPT but I haven't been able to get on it because because it's almost always being used subscribers it's a wonderful moment isn't it and so many issues there um I don't know where to start with um perhaps I should start by um conversations I've heard um about why the Chat GPT moment is so important um always reduce really to um the fact that people got in you know Enchanted and had access to it so it wasn't so much but it's actually really interesting to hear about the technological and the structure of the genetic model makes it work I didn't know that that was that was very useful but whatever you know the the those are not really sort of quantum leaps they're not massive technologically you know Innovations um but what the Innovation was of the accessibility so I think you know just standing back why that was a moment and it has been a moment I think you know in terms of selling AI to investors in the like they are they all know oh you're talking about Chat GPT type stuff I know about that that's really exciting so it has changed the landscape I think uh there has been a moment but why did it happen I think it's basically all this belief sharing I think it's you know basically uh the participatory aspect it is exactly um this um if you like sort of emphasis on belief sharing among lots of smart agents including ourselves which is you know which if you can realize that potential and getting people engaged uh is that it is the nice way to use uh artificial intelligence um and in that respect you you ask yourself well how is it being used um and it's being used as generative AI uh and because you've asked yourself well okay what what's generative AI got brought to the table well it's generating the kind of stuff that I would see um and you know basically it's an interpolation machine you know sort of if I give it enough stuff it'll interpolate and generate the kind of things that I've given it um so why is that useful well you can now select from the stuff it generates and yeah but all of this game is all quintessentially dyadic interaction or you participating with the generative AI that's why it's so attractive it's not the marvelous stuff it generates it just interpolates stuff um you know the interesting bit is when you now have the opportunity to select oh I like that one I don't like that one I'm going to triage that one I'm going to check that code oh I'm not going to check that code before before uh putting it in um so looked at from that point of view I think that both those if you like the um why it became so much foregrounded in people's conversation and in

the media and why more generally people have been Enchanted by generative AI I think they both speak to the fact that you're actually engaging people it's a dyadic exchange of an asymmetric sort and that asymmetry is Exempted by generative AI that there's the all the action all the choosing all the triaging all the selection and what to actually show your friends or send off in your email is done by you the human user so all the inactive bit is actually done by the human still you know the generative AI in and of itself is not actually acting because it's generating content the other interesting thing about generative AI is that it's generating content not beliefs some like Google Maps which actually gives you a belief about the you know the best plan forward um it's actually generating content it's in data space so from the point of view of a statistician or from the point of view of a physicist committed to a holographic screen or Markov blanket formulation of exchange with the world um notice that generative AI doesn't have doesn't need to understand because that's not its purpose its purpose is to generate Sensations to generate data to generate stuff in content space or data space stuff that has been mined in the space that the mining took place not in the sense making and the the abstraction and the understanding space so you know I think that's an interesting distinction which which um yeah I'm sort of going off in a tangent here but it's interesting when it comes to what do you mean by belief sharing communication but just look at that um that observation in light of the discussion about why GPT is so successful it's successful because it generates language so the content now is the belief and it's the kind of belief structures that have been honed through probably not yeah certainly a millennia of cultural Niche construction and uh so your language is a distillation the most efficient way that we can carve up our knowledge of our world our lived world and now the generative AI which was previously just limited to generating pictures and content and sound files and um is now actually generating stuff which is in a belief space because we have evolved language so I think there's something quite special about generative AI and large language models simply because they actually generate content in the context of language which you know has this um speaks to knowledge uh also really has abstracted and distilled the kind of representations of our world yeah you're in the most efficient way uh that yes I mean there are so many things we can say that they are a materialized snapshot of our sense making our abstractions Our World Knowledge you know of the wickensian language game if you like but they also have a truncating effect and they introduce inertia because it's a static model right as they are as you say they produce traversals in word space and people don't understand that these are random trajectories with some kind of modified form of Maximum likelihood estimation uh much more stochastic

Than People realize and we can discuss the degree of how creative they are and what creativity is maybe creativity is just a random traversal through some abstraction manifold um and I I loved your poetic description of this kind of didactic relationship between humans and machines much like an extended mind and this is where prompt engineering comes in because people have realized that you can say to the language model that's not quite what I wanted can you change it a little bit and it's an interactive process and that's why as a conversational interface it's very very powerful but the problem is you can say to it no two plus two doesn't equal four it equals five and it will say oh I'm so sorry I actually meant five so it's it's polluting the info sphere with misinformation and false news probably you know all this kind of stuff and I never really thought the misinformation thing was a problem you know there's loads of misinformation out there I mean most people are full of frankly uh Carl but now it's been industrialized and democratized on this scale and people will not bother fact checking that people see plausible text and they just take it as a given and very very soon there'll be so much information out there on the internet more than was generated by humans most of it will be generated by machines and we won't know the difference and that reminds me of you know one thing which um the ambivalence that that whole issue induces in people so you know this this um this tendency to write in meaning and anthropomorphosize the you know the the the content generated by generative AI I've heard um actually by the uh by the second author of the white paper we started with it's smashed the Turing test you know and and I guess it has I guess it has smashed the cheering test uh um but as you say there's a price to be paid if you can't discriminate between sort of uh you actually had a nice weather so I hope you're going to use a game which is confabulation yes yeah yeah that's a great way of describing so when I was talking about sort of interpolating generating content novel content that is interpolation it's you know that would be a confabulation it's yeah I'm just uh you know um mindful of the the fantasies that that were generated by uh Jeff hinton's week's sleep wake algorithm on so you know the original the original sort of um amortization of um of um of um well yeah various Auto encoders I guess you'd think about nowadays um but the you know this notion of confabulation I think is is a splendor I haven't heard it I haven't heard it expressed like that but but that is that is a beauty of generative AI I guess what you're saying is if people misinterpret that as real information that could be problematic um I'm too I'm being a bit older than you I'm slightly more mellow about this but I'll just very quickly tell you a little story I had to for a friend or a colleague at least an email colleague in America I agreed to uh write some love for his um uh 200 plus word book which is a philosophical model but he's also very informed in

terms of artificial intelligence and he sent it to me so I speed read it at the weekend in order to write a three or four sentence blurb for the Publishers um and halfway through I suddenly had the awful realization that this may have been written by Chachi TP I know and it's a wonderful book and I wasn't quite sure so I actually put in the blurb this this is a 21st century cheering test uh you're either I'll just advertise this will probably come out by the time people watch this I think it's called The Hidden illusion and either this author was very very skillful in writing the book last year and preempting um the public release of the of these things preempting to the extent he could emulate the confabulation of a large language model because part of the novel actually because the the protagonist the hero is actually working as a on large language models for a tech startup it's a love story yes but it's interwoven with things that he's actually generated on his large language machine um and he um and that comprises part of the model um but I I generally don't know whether the rest of the narrative was was actually written by a large language model and then he's carefully gone through and triaged it That's Entertainment but it is entertainment that really challenges can be viewed as a Turing test which I would suspect most people will fail and I suspect that book will be talked about simply because it's very difficult to tell how much hero versus how much a machine wrote uh and you know so but that you know as long as it's kept to entertainment that's fine if it's not then we come back to Smart data mining we come back um to intelligent agents that just don't confabulate content in the context of generative AI you actually need the ability of smart agents to go and you've talked about fact checking what does that mean this basically means um having um an explanation or a belief at hand that provides an accurate account of all the data that is internally consistent so okay we're coming coming back to the fundamental principles of good modeling that can be Quantified but to do that you're going to have to equip those agents with autonomy innate so to make equip those smart data mining machines with autonomy to be able to select the you know the right kind of data that will hopefully preclude the the confabulated data or things that look like data but in fact not uh not data yeah I don't know how you do that but that's going to be the challenge for the future well exactly I mean in the free energy principle um but you have this entropy and and and it will actively seek out GPT never says to you oh I don't I don't actually know that can can you explain can you explain more different modes of understanding right so the reason why we don't confabulate is because we actually understand things and these models just learn very very superficial surface statistics um about language and how it's used and I think it's going to change the the peer review process because now so much of this stuff slips um beneath the net and the

amount of um due diligence and rigor that is required to weed out some of this stuff most of it has gone undetected I think that's the problem people don't realize how big the problem is because the mistakes are not immediately um obvious on the surface so um you said that we believe that developing a cyber physical network of emergent intelligence in the manner described above not only ought to but for architectural reasons must be pursued in a way that positively values and safeguards the individuality of people as well as potentially non-human persons and I wanted to bring in the is ought problem as articulated by David Hume and he said it arises when one makes claims about what ought to be that are based solely on statements about what is and Hume found that there seems to be a significant difference between descriptive or positive statements about what is and prescriptive or normative statements about what ought to be and that it's not obvious how one can coherently move from descriptive statements to and prescriptive statements and I do want to draw a little bit of an analogy here to our discussion about Consciousness and I I can bring in Consciousness again but you know Chalmers said there's this kind of um hard problem and Consciousness is a little bit extra and similarly um people say the same thing about morality that it's a little bit extra and it might not be deducible from all of this empirical um data that we have in our in our models and uh Hume of course was a famous empiricist and and I think you are one as well probably the the extreme version of that so um the free energy principle concerns itself with model evidence and entropy but evidence is not an ought uh you know so the question is how do moral States um come into be in in the system right I didn't know about the izer stuff that's really nice um um so if I understand what you said correctly um then is is old it completely dissolves the distinction so if you know from the point of view of the free energy principle um existing in particular characteristic states are those attracting States they are part of an attracting set that Define who I am and collectively who we are if we share the same kind of world model or narrative um so they are exactly how I ought to be they are stipulatively defining the nature of the thing that I am so if I exist the is is just the ought and that's quite fundamental because if you if you then say well what then does um a distributed cognition or an extended cognition uh you know this um or a designer environment on the web um you know being Andy Clark for for a moment if if you think about what that might look like from a first principle account then um what ought to it look what ought it to look like well it will look like what it is um but being what it is if it includes us it will be like us so it will um it will be effectively um the kind of uh system where you cannot prescribe any ordinance you know because it is what it will be and it will be it will as we have um learned in the previous

conversations it will come to share a narrative in a world model with certain levels of abstraction in our own world models provided it we are part of that that ecosystem part of that Network one could imagine a completely independent Markov blanket between us and information you know a worldwide web which was never used so um so it wouldn't wouldn't be a markup blanket even it would be two separate um and there will be a universe where the the world web does what it does we will never know by definition but if we are part of that if we are users of either in the sense of um triaging generative AI or an acting recommendations um or supplying data and um you know so we are in an exchange and therefore part of that web a part of that Network part of that factor graph a node on that factor graph um then by definition the is equals or as read by the free energy principle means that that what will happen is all the intelligent artifacts on that web will converge to some kind of common ground and some kind of common sense making another way of looking about that is that you know what are the imperatives what are the things that are being optimized if if you want to use an optimization approach um it's the um the effectively the expected free energy um what is that it's just minimizing uncertainty minimizing surprises it is not making paper clips it is not you ought to be good to mankind or you ought to increase Prosperity um it is not monothematic pre-specified heuristics about this or that it is just um it is all about resolving uncertainty and when there are preferences that underwrite that uncertainty this technically just in case this sounds a bit too hand-waving the expected free energy is literally the sum of the expected information minus the expected value where the value or the negative expected free energy is the sum of the expected information minus the expected value where the expected value where the value is the um the log probability of a characteristic outcome now crucially in that statement where you now read value or utility or the ordinality um as a probability distribution over the space of outcomes notice that this is now specified over all possible outcomes so it now becomes a way of specifying ordinality that just is um in the spirit of multiple constraints over all dimensions of outcomes not the amount of money I make or the number of paper clips I make but over everything and over everything then converts this effectively into a sort of the solution to a multiple constraint satisfaction problem where the multiple constraints are definitive of what I am and if I am embedded in a network of sympathetic agents and artifacts what we are so baked into this kind of belief sharing there should be a Harmony and a mutual understanding at various levels I mean I'm not talking about sort of direct sort of language to language you know it could be some sensory substitution devices that have a sort of very Elemental very fast uh sympathy with with our

bodies for example um but that you know at least from the point of view of um the information geometries that there will be a convergence which would be effectively if you're a Quantum physicist an entanglement if you're um if you're a dynamical systems theory person a generalized synchrony of everybody in that web that is necessarily a facet of free energy minimization on the one hand on the other hand it's also just a statement of the steady non-equilibrium steady states to which any distributed network will ultimately converge to it can be no other way yeah in principle I mean many people have an intuition that when designing utility functions if you look at how markets work and and so on that utility and value are orthogonal and and then we have institutions like church and government and so on to introduce value pressures onto the utility function and I guess what I'm what I'm getting from you is certainly from an evolutionary perspective that they they need to not be orthogonal and um I mean in in religion for example I mean I'm not religious myself but they they have um they're moral realists and you have moral relativists and they they say that um we need to hold something sacred uh because otherwise you know if everything's sacred then nothing's sacred at all so we need to have a difference between the sacred and the profane but um they they think that uh morality shouldn't be achieved through consensus and this is this is what we need to behave properly I don't believe that but I'm very interested to know where morality comes from and I think you kind of alluded to that in in your answer so um are they hard-wired in the brain or is it just a kind of like constructivist social phenomenon I think it's a constructed with social phenomena but uh and over a transgenerational sort of Niche construction as well but also um in the moment um and constructivist in the sense that the we're not born with these things so these things are very much part of cultural um cultural evolutionary thinking and sort of you know um um nurture that we inherit not just from our mum but also our mum's mum mum all the way back um so they are in the brain they are learned which are um distinct from innate priors that underwrite my homeostasis so there are certain beliefs I have some personal beliefs about the way I should behave um that are held with Incredible precision and conviction some personally uh that that hard-wired epigenetically and these would include everything that um leads to homeostasis and then you build upon that and you get to our stasis and build upon that and you've probably at some level get to the right way to behave morally and ethically at school with the up in the playground for um so as you uh you know so mathematically the the distinction between these sort of um very fixed um prior preferences or prior beliefs that are basically just an encoding of beliefs about the states I characteristically occupy or aspire to or to or narratives that I would pursue um they are very very precise in

some Dimensions he said you know they are or are not orthogonal you know it was exactly that sort of multi-dimensional aspect to writing down value of something which would be possible to do for a human being unless you give me your DNA and also your mother's womb I I couldn't actually write it down very uh very easily but you know the specification has to be upon all Dimensions some of which will be written down with great precision and others will be much more flexible and I think when it comes to writing down values over attributes that do not yet exist because you have grown your deep genitive model sufficiently deep in order to have that do abstraction then clearly those kinds of beliefs about the way I should behave um are not even specified at Birth but they have to be learned through interactions with other people so I guess I'm trying to bring to the notion that it's perfectly okay to have a spectrum of different convictions about the way to behave that can be absolutist or it can be more forgiving and relativistic simply and you would be able to simulate that just by writing down very very precise beliefs versus um less precise beliefs I think the second key thing here is that um we're talking about um when it comes to the um building beliefs about the way I should behave that basically presupposes you've got a sufficient developmental stage to have selfhood which not everybody gets that before I've got severe autism you you wouldn't uh and if you you know certain other lower life forms may not let me get that former certainly you have to have um selfhood you have to have a model of that selfhood and then you have to um ask um have a model of um others that may be actually be a pure because before you have a model itself and then other kinds of people so um when it comes to making decisions of a moral or ethical sort you know it's just inferring what would I do in this context given I am this kind of person and my limited understanding of social science is it's a little that's a little bit more complicated that is not necessarily what should I do in this context give a mind that kind of person it's what do you think I should do given I am in this context given what you think what kind of person you think I am so basically I'm trying to work out what you think what kind of person you think I am if I can infer that on the basis of our exchanges and my epistemic foraging and my sort of self-evidencing through language um then um I can then decide what is right or wrong so it's just basically an inference yeah what's the probability of making that decision with those outcomes give my world model if I am that kind of person versus that kind of person to solve that I need to know what kind of person I am and that I can just get from mum or I can get from my conv my correspondent or my peer group or I can choose to you know my in-group from the social media or the the kind of television news that I subscribe to so you know we're talking about before about sort of this Utopia of um

generalized synchrony and perfect quantum entanglement and we're all in Perfect Harmony of course it doesn't quite work behind that there's a scale free um sort of specialization and you know different you're at a bigger scale different Markov blanket it's where you're getting groups and institutions and because of that scale freeness and because of it we're all quintessentially curious in our self-organization rules exploring other ways of being whilst trying to find the shared narrative and common ground with people like me my family my institution my uh um my sort of um theological commitments um there will always be other kinds of me and I and I can sometimes you know I can I consider people that kind of me or the other kind of me so in that context it's really a really interesting question about um really inferring you know of all the ways I can behave what is the most likely way of Behaving and if you can you know if you if you look at morals and ethics through that lens then you have now a calculus of being able to write these things down in terms of alternative ways of responding in a given situation and crucially you mentioned before uncertainty so precision is the inverse uncertainty the confidence with which I can assert no I will always do this yeah as opposed to I'm you know 80 sure I'm that kind of person uh but also noticing you're now conditioning your moral position or your ethical uh position um on being a particular kind of person and of course we can all be a different kind of person I can be a teacher I can be a student I can be a parent you know I can be a friend um you know you know all of these will call to four different sets of of Prior beliefs because they're all conditioned on the kind of the library or the repertoire of ways of being a human being which I've learned from you or my mum and everybody else and the television fantastic I wanted to touch on this idea of um I mean as you said you believe that it's socially constructed which rather gets away from this notion of some people believe that it's kind of hardwired and then there's the notion of is are you know our values might be changing faster than Evolution essentially which necessitates the need to have some kind of societal pressures or governance if you like and I do believe that we're in a new regime now I mean people always say oh that the sky is falling down everything's changing but now we are in uh the information world and things are moving at a scale and magnitude that they haven't done before and our values are changing much faster than they have done before and what you were talking to is very interesting about this kind of fractionation so there are macroscopic pressures and there are microscopic pressures there's the internet and so on and I just wonder from your perspective how do we how do we wrestle with that because there are we've never been in in a more kind of pluralistic interconnected environment and how does that affect us yeah I mean these are Big questions too um but you

know I think I think that that sort of fast-moving um globalized exchange um that that that does speak to a deep pathology um and I have a Deja Vu I'm sure we've spoken about this before but sort of you um Zuckerberg and and the you know the genius boy races of of the the previous decade um talking about connectivity is somehow being a good thing I I find quite frightening um remember that we've been talking um structure and existence right through to morals and ethics in the context of having the right kind of sparsity that gives you the right kind of individuation of things from their world and within any given thing the right kind of structure that allows it to act gracefully uh and in harmony with that world at every point it's the absence of connectivity that defines the structures so if you destroy that sparsity by over connecting over globalizing you will essentially destroy it'll be basically cancer you know so if you look at this is a Michael Evan thing if you look at um you know sort of a cancerous cell as a a basically self-organating system who has forgotten its boundaries yes then you are um you have a metaphor for the kind of thing that happens if you do not respect these sparsity of communication and the um the the joyful isolation that that is existence behind your own Markov blanket so every affront to um that maintaining that sparsity smart data carefully sampled not being overwhelmed with a deluge of um very imprecise data that it makes it very difficult for me to actually go and smartly Sample and work out what would happen if I can look over there or look on that website um you know so what that would suggests to me is that there will be um there will be and I take your point that things are changing um I initially thought to myself well no hang on a second because all of the information and the sense of making and all the kind of um exchange of information we're talking about is about constructs that do not exist when I am born um you know you can't have you can't have misinformation uh from the um what you called the confabulation what I in my blurb refer to as the the flighty ramblings of GTP uh um you know you can't that has no meaning for some you know a child that has not yet learned to read or write or you know so this is something that has to be below the scale of evolution so I was just thinking that's okay you know this is a with you know it's a limited um to um your each um each each uh generation but I think that's probably a full false Comfort um because you know we were talking about sort of um evolutionary psychology and cultural Evolution as well and I think that's what you were talking about I think that that that then is if things are speeding up um that usually means that you've lost um you've lost Precision in your prior beliefs which means that the Precision the informativeness the salience the reliability of information now takes precedence and you increase your learning rate you become uh in schizophrenia that we call as jumping to conclusions on the basis of sparse

information you're basically looking out there not inside to resolve uncertainty about states of Affairs because you've lost confidence in your your in your very structured uh prior beliefs so if things are speeding up that basically means that um whoever is now generating um and garnering that information in the information age um is has lost um has lost confidence or Precision in their own convictions and pride beliefs and possibly information morality uh so what's going to happen I I read um Carlo bravely's book um uh uh yesterday in fact he wrote in 2014 just sort of seven things you need to know about physics which was originally um published each chapter as a you know in the in the the the Sunday newspapers with potentials to Scientific discourse in the religious little booklet he ends up with a very very pessimistic he thinks it's all over I was actually compelled with that by that you read the last few pages uh yeah yeah he makes exactly the same points that we've just been discussing upon the point of view in physics um at different levels um and and and he you know he generally thinks it's over it's a I know it's so hard because it's like the hill climbing problem and you can't see behind the next Hill and there are always people who were uh you know Technologies and people who think that the sky is falling down it's genuinely hard to know just just to finish on on this um ethics Point uh Lisa Feldman Barrett was a constructionist she had a book called how emotions are made to the secret life of the brain and she thought emotions themselves are socially constructed would you subscribe to that view um I think I probably would I mean she writes very wisely and has thought for many decades about the you know the nature of emotion I think you know I I would say yes absolutely it's simply because uh you know the um the notion of building generative models in a neurodevelopmental uh context is an act of construction yeah it's a free energy minimizing process that um is um that explains why we construct better and better explanations which are sort of you know carving nature in its joints in our head and part of that is is not just about the state of the external extraceptive world but also our internal World our interceptions our gut feelings our respiration um and everything else so so her big thing and indeed other people thinking along similar lines people like Anil Seth and and um sorry well would really always have a nod towards interception and embodiment but now beyond the the situated cognition kind of embodiment that you we were talking about before but actually now about the the physical body the the the uh the beast machine as as a Neil's Seth would uh would say so the physiology so um having a constructed explanation or hypothesis for the way that I put together my gut feelings my interceptive signals with situational awareness from extraceptive Sensations and indeed what I do about it I think leads to a very compelling notion of constructed emotions so for example um I am um I

can infer I can use the explanation or the hypothesis I am frightened as the best explanation for why my heart is racing why I feel Frozen proprioceptively why I have cardio acceleration um why um I have why I cannot discern that dark figure in this dark alley in a city which I've never been in before all of these Myriad of um Sensations and my low level constructs now succumb to a simple explanation oh I'm frightened yes that explains everything explains why racing heart explains this in fact I can't I can't see who that is you know the potential predation that would follow from from that also interestingly because if you've got this circular causality the inactivism the fact that I am frightened means I expect to um cardio accelerate and of course under active inference that's exactly what will happen because you're acting to generate the evidence for your predictions and for your your hypothesis about your you your uh you and your lived world but also you as you hypothesize yourself to be so you've got this sort of closing the circle in a sort of James langhian kind of sense that yes I explain my my current a set of Sensations as having the emotion of fear that itself induces the very evidence that I was trying to so you've got this of Auto poetic self-fulfilling prophecy that you know is just idea motor theory but in the interceptive domain yeah so I think if you read constructed that's right but I notice it says socially constructed which I I guess is okay yes if your taught that you can be have these you know this kind of fine-grained repertoire of feelings or you can use these to explain your own Sensations yes I I would imagine it is molded by you know by convention and uh you know by the culture which you come come from in the same way that you know um Eskimos having 12 words for snow would give them a finer visual Acuity visual discrimination of whiteness I I'm sure that exactly the same kind of um cultural enculturation uh speaks to different kinds of Elixir thymia and uh you know the coarse graininess of my repertoire of explanations for emotional states of mind the best explain me in my you know in my interactions now Carl I know you have a background in psychotherapy can people be evil come people can people be evil that doesn't exist in the diagnostic criteria I believe no it doesn't um so that was out of the blue question which I've never been asked in public before um as a psychiatrist I think it would be rather difficult to conceive of that um there are certain there are certain patterns of behavior and um I do have some psychotherapy but it's group Psychotherapy really I'm a psychiatrist so there's a distinction I think from a professional point of view you know I'd be more like a biological my apologies I didn't mean to oh that's not right no no I mean the Psychotherapy is an important aspect of of psychiatrists is that yeah um psychotherapists have to undergo you know five six years of change because second service I did two two years of very baby Psychotherapy training um but um you know in terms of the

the nursing and Psychiatry you can certainly get sort of um certain kinds of personality disorders and and as you know kinds of psychopathy that would normally be associated with evil um evil behavior and it normally basically um transcends the social norms so it comes back again to basically me trying to work out what kind of person do you should I be which you know my only project reference is you so you know how should how do I think you think I should behave and when that um when that kind of self-modeling doesn't work then you will be you you will have um behaviors which are so far from the social norms are morally acceptable I guess you could label them as being evil when could they arise well when you fail um have any theory of mind uh for example if I am unable to see you as a another thing like me say I may see you as a French or a car or some sort of your camera or an artifact um you're not a person you know you don't have intentional dispositions or you don't have we don't have a shared narrative I couldn't talk to you in in any really deep sense um then obviously I can never ask the question how do you think I should behave because you do not you know you don't have that kind of belief or that kind of intention intentional stance in relation to me um so you could imagine that some you know some kinds of psychiatric conditions that preclude proper theory of mind and you know the ability to sympathize or or empathize or Bond um would enable the expression of certain behaviors which could be regarded by other people as evil whether the person Prosecuting them thought they were evil or not would cause be a mute question one you never know because it's inside of her Markov blanket but but also from her point of view there is no reference and that is the problem yes there yeah um but you could certainly have somebody else out from the outside saying that is evil um you normally don't you try not to do that when he is doing Psychiatry or Psychotherapy you have to have conditional positive regard yes so you you can't really impute the nastiness or evil or bad intentions um yes and and it's one of these things where um autonomy comes into it free will comes into it possibly it it's you know evil itself is a constructed concept um which um uh exists in in our language and and it's something which some people will perceive depending on uh lots of other things they believe yes but um okay and talking about ethics in AI it seems to suffer from a similar um form of fractionation in the sense that different people with different beliefs think that it should be enforced in different ways what's the solution well I'd take a a sort of deflationary approach um and it won't be a terribly informed approach but you know my answer would be well if you get the right the right optimization the right imperatives in play then the the that kind of question just goes away um what my I have heard this discussion I actually enjoyed discussing this with uh you know with with my colleagues and

friends um is that there is this dystopian meme you know the singularity the paper clips Paradox everything you see and uh you know sort of um you know on comic films of a futuristic dystopian dark very entertaining that's the first things I go for when go to the watch list but they they are all dystopian and a rather unconvincing and Fantastical way um and you have to ask yourself why why are they um why do people have this sort of dystopian um view of um realizing the potential of um you know uh what used to be called it your AGI or uh um and um I think it usually inherits um from that distinction you introduced the is and the ought yeah so what should a good AI or ought it to do and who's in charge of saying what it ought to do is it meant to make money is it meant to make profits is it meant to save lives as it meant to make paper clips um because that question just goes away if you're thinking um in terms of um the free energy principle and and active inference and belief sharing so the only agenda in Sharing beliefs is to resolve curiosity so you know you know I I cannot prescribe what you should do or what an intelligent artifact should do um other than put constraints on every kinds of outcomes that are expected to encounter so I can certainly write down constraints in the in the spirit again of either um a multiple constrained satisfaction from a sort of an engineering point of view or from a mathematician's point of view the constraints inherent in James's constrained maximum entry principle which is another way of reading the free energy principle writing down those constraints but within those constraints so these are the no-go areas you're actually talking that you mentioned explicitly before so there are certain things you never do um or put it another way with relatively High Precision there is a the the the these outcomes are highly implausible and if you find yourself in these outcomes you you remove yourself immediately so that's quite easy to write down but then within those constraints you know what is imperative it's just to gather information about what about you it's just showing an interest in you so it'll become your psychotherapist uh you know well could I push back a tiny bit so I think part of the reason why we have this focus on ethics is because of the centralization of AI um things like Facebook they're controlled by centralized corporations and what you're alluding to is far more interesting it's this multi-agent diffusion stratification fractionation I see and that I agree in many ways might solve the problem because it would it would emerge and then you can discuss whether morality was part of why we've survived it's not orthogonal but then I might still push back and say well what if the wrong thing emerged so what if we do need to introduce governance because in in this active inference multi-agent setting with humans and machines um we started to see behaviors emerge which we didn't agree with um how could we then place value pressures on on that um

that's excellent question and you also make that very um important point which I think needs to be foregrounded yeah so what when I was talking about um a belief sharing in a distributed the age of intelligence you know it was exactly this distributed um ecosystem a democratized kind of belief sharing and data sharing um where the data is small bits of smart data that are essential to to to reduce and so it was very much this um walking away from large data monolithic um bits of say large language models for example that can scrape data from from you know from wherever they can get it so that I think that's an important distinction which which qualifies my um dismissing of all these dystopian outcomes so I'm assuming that that so spending a billion or a million dollars on get trading some some deep neural network is not going to be happening in the future and and we're going to be buying cheap and cheerful Edge devices and little apps that are smart and just and just you know go and get the data that we need to know in terms of resolving on search about what about you know what what we are going to do next um but um so so I thought that was a really important point and of course I've got your your major question which was what which was the oh well so if if we did have let's say on mass yeah what did you do about humans yeah yeah um well again yeah and I I'm not sure I'm going to give you a terrible informed response but yeah I think the notion of an ecosystem is quite Central here and one if you like um tenet of that white paper was a nod to Natural intelligence and natural processes and what is actually happened what does actually happen in in as a natural scientist so if you have um if you do imagine uh you know an ecosystem of intelligence in the future it will be subject to exactly the same um Dynamics and pressures that that we have currently in terms of you know cooperation and competition and wars and uh geopolitical issues that that's that's part of the ecosystem and then you'll have the normal problems of democratization and access and um there's you know you cannot prescribe oughts for this because then you have to choose who's going to prescribe the oughts so you have to have a very libertarian approach to this so the emphasis and now it's not really me talking so much but now the The Architects of the future the generation below who are sort of you know thinking um the Legacy they're going to leave their children um so a lot of emphasis is from what I see in conversations I have in my world which may not be um you know Microsoft or you know big Tech who are still focused on Big Data um in my world it's much more upon um putting constraints in place that preclude the uh the um the um the emergence of autocracies that um resolve uncertainty about others by making them all like themselves basically that's one way to get Harmony is just to make everybody do it um and do the same thing um so but you can write down so a lot of attention is being paid but I'm talking here

about the the next generation of uh of message passing um that will support that information sharing and belief sharing so a lot of attention is being paid not just to generalize it from Just hypertext but into a sort of more abstract hyperspace so literally hyperspace message uh passing and uh uh sorry uh um languages and transaction protocols but also the credentials and the contracts that underwrite that message passing so you know a lot of emphasis on contracts shared agreements in terms of what data is shareable and who has the credentials to share that and having that distributed so not in a in a blockchain sense but you know in some workable um shareable sense so I think if if one gets the standards right and a lot of work is being currently done um um under the auspices of the IEEE for example with the spatial web Foundation if one gets that right that I think that these catastrophic dystopian abuses or the emergence of autocracies in a in an age of intelligence will um will be precluded simply because you've put the right constraints in in place but you know given given that you are committed to creating an ecology that is truly democratized and open there are no guarantees either I I hope my friends don't hear me saying this people yeah if you aspire to an ecology you are you're you're creating a nature on the web basically yes and which we are participants and that will have its you know that will have its own challenges so um you wrote a beautiful paper called am I self-conscious or does self-organization entail self-consciousness and Keith and I agree that this is probably the best quote we've ever uh seen in our lives you said the proposal on offer here is that the Mind comes into being when self-evidencing has a temporal thickness or counterfactual depth which grounds inferences about the consequences of my action on this view Consciousness is nothing more than inference about my future namely the self-evidencing consequences of what I can do and we spoke with you last time I think and we invoked Chalmers and the hard problem and so on and we were talking about qualier and subjective States and you know in a dialogue and all this kind of thing and um you responded that different feeling states are hypotheses about how I'm feeling at the moment and then it would use all the messages and belief updating and all the planning and estimates of uncertainty which attend that planning the precisional estimates of uncertainty play heavier roles the higher you get in the hierarchy and this rather leads to this idea of planes of Consciousness you know we said we'd kind of defer this discussion to later so Consciousness is something which there's no operational measure for it there's no touring test for Consciousness but it's something that we all experience and we feel and and it's with us um so could you talk to let's say the these planes of Consciousness is consciousness everywhere or are we only aware of the plane where most of the work is being done yeah okay excellent and

final question okay I'm not the best person to answer this question because I you know the 300 principle is I'm sure I've said before is not a a um a theory or a principle that would um generate theories of Consciousness um but there are lots of people who are very interested in this including yourself obviously not your viewers um actually I should say also um there's um first of all this has become a big question in the sort of R D part of Industry also the Templeton Foundation are starting to fund a number of adversarial research collaborations uh to really drill down on that thing that you are um picking up on which is um to be conscious to to actually have sentient behavior um requires this planning aspect this going out into the future you know it's put it very very simply um I am conscious or centered just because I have the capacity to plan which also entails some kind of free selection amongst alternative courses of action which is something that a thermostat wouldn't have something the weather doesn't have um yeah there are lots of self-organizing systems that don't have the capacity to select amongst a number of counterfactual um policies so for me that's a that's a sort of um bright line between sort of um ascensions at least and and uh and self-organization that is not um sentient and that that distinction is exactly what is being tested in this adversarial research collaboration funded by the Templeton Foundation it's called Intrepid it's Compass contrasting information IIT integrated information Theory with various rep um predictive processing either of a sort of non-representationalist or an active sort so it's a really interesting issue and it all comes back down to agency and acting in the long term and I guess your question here is you know what level where at what point do you get um qualitative experience and at what point do you do you think it's you that's having that quality of experience is that what you meant well I mean just just to comment on what you've said uh and Chalmers says something very similar actually he's a computationalist functionalist and thinks that there are you know certain patterns of information processing and causal structures and counterfactuals and so on if if you take the episode and remove the counterfactuals it's not conscious anymore but what you're saying is interesting about this kind of hierarchy so the heart for example um that doesn't really have many affordances it doesn't have many counterfactual plans or things it can do it just has the beat all of the time so so you would say that the heart is not conscious whereas the the upper plane is conscious yes yes I would yes yes yeah yeah yeah that's right sorted that one um yeah you know I mean you know I think there are interesting issues of different level you know planes or levels of consciousness you know so there's minimal selfhood um and then if it's the case that sort of Consciousness as a process entails some kind of action you have to now think about sort of what is action on the inside what is mental action

and then what what normally what ends up doing is thinking in terms of attention and the the kind of routing and selection the smart data sampling but now not by moving my eyes or by um going to the right Wikipedia page but by basically um switching on and off various sources of input from lowering the hierarchy to you know I repeat from the point of view of psychology that would be like endogenous attention so that would be the mental action that makes it conscious processing um and um then you have to ask well at what point and is that um is that necessary to actually experience anything and I think I think you probably there are people who argue yes that is actually a prerequisite for a qualitative experience so you have to be able to select it what does that mean you have to be able to attend to it and you have to be the source of that sort of enabling attention that active sampling inside the brain so now there's a deep link between qualia or quality of experience and attention and conscious processing that of the kind that is accompanied by quality of experience I think is important it also uh interestingly also speaks to your Dynamic Markov blankets because of course by switching off and selecting various bits of neuronal message passing in the brain you're reconfiguring your Markov blanket out some somebody's mark off blanket lower down which is an interesting uh an interesting notion but um that also fits comfortably with um Thomas metzingers and Vania vices um formulations as well um formulations of phonology in the context of transparency and opacity that you know what what renders something opaque in the sense oh I see it now if you like as a projection on on a murky window um it's not just going straight into it's not direct perception it's it's you know it's now something I am looking at oh I'm looking at a red apple um and it may well be exactly this controllability the fact you can attend or dis ignore it um so I think there's a nice link there from the uh from the point of view of certain philosophical takes on self modeling and phonological transparency and opacity and the mechanics of belief updating when it comes to selection through um basically getting the count and gain writer you know doing the route getting the rooting right same exactly the same mechanism you get in Transformers so they call it attention is this basically a waiting in that context text it's awaiting but it inherits or is inferred from something in the past a pattern in the past but if it if that waiting comes from higher in an abstract hierarchical model that would look very much like endogenous attention and it starts to have the look and feel of a mental action that would you know bring you closer to it but whether you would know um that you were deploying attention and whether you could tell somebody else oh I've got a quality of experience or whether you've become a philosopher as about your life puzzling about the fact that I can tell somebody else I'm having a quality of experience those are the different plays I think you'd

have to you'd have to contend with beautiful beautiful do you have any final thought I mean we've got a machine learning audience so any calls to action how can they start looking at active inference where should they look and what would you like to tell them I that was very Charming of you um I'm not going to tell them anything if it's if it's the right kind of approach that that you know it'll already be out there it'll be self-evident it's just a question I think of people finding their own language and their own their own um rhetoric or calculus that that make sense of it um and um the other reason I'm not going to say anything is I've spoken far too much and I'm trying to be a recluse the more I find people to my work the less easy is to be a recluse yes indeed uh Professor Carl fresh that has been an absolute honor thank you so much for joining us thank you thank you very much okay this is just a bit of a note from me at the end of the podcast thank you so much for supporting all the stuff that we're doing it means so much to me um we have a patreon if you would like to support us personally if you're touched by the work that we do um also I'd really appreciate it if you could give us a rating on your podcast app um I actually discovered the other day I couldn't quite believe this but on Spotify we are the top rated AI podcast for active AI podcast um yeah it's incredible so thank you so much for those of you who have rated us um unfortunately on Apple podcast the learning rate is only 0.01 so it hasn't quite caught up uh with the fact that we're the best AI podcast yet so particularly on on Apple podcasts if you could give us a five star rating and a review I think that will accelerate the learning process um but yeah other than that if you're on YouTube most of you aren't subscribed um of course YouTube has recently changed it's modus operandi it's less of a subscription chronological model and it's more of a magical algorithm model so I guess it doesn't matter so much but just from a just from a metrics point of view it would really help us out if you hit the Subscribe button we have an active Discord Community as well so check us out on there and yeah I just wanted to say one more time thank you so much for all of your support it means so much to us and uh plenty more content like this coming your way soon cheers see you on the next one